Machine Learning of 3D Scene Geometry
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MASTER'S THESIS

Foundation-Model-Guided Zero-Shot Synthesis of Human-Scene Interaction

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Declaration

I declare that this thesis has been completed independently and that no sources or aids other than those listed have been used.

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Abstract

Generating a human who interacts convincingly with a real 3D scene is difficult because the input is usually semantic, while the output must satisfy geometry. A hand should meet the correct object surface, feet should support the body, and the body should not pass through nearby furniture or the room itself. This thesis studies a zero-shot pipeline for static human-scene interaction synthesis from a single RGB view, a ScanNet++ scene mesh, and a language instruction.

The method uses foundation models where their strengths are most useful, but keeps the final result tied to metric 3D geometry. First, a vision-language model converts the instruction into a Scene Interaction Graph that names the target object, relevant human body parts, and intended contact edges. An image editing model then inserts a plausible human into the scene, producing a 2D reference for the interaction. A second contact-inpainting step estimates object-surface contact masks for the body parts specified by the graph. These masks are projected from the image view onto the visible scene mesh and converted into 3D contact targets. Starting from an SMPL-X body initialized with GVHMR, the final stage optimizes pose, orientation, translation, and scale using contact, depth ordering, scene collision, self-intersection, height, and pose-prior terms.

The pipeline is evaluated on real indoor scenes using physical contact distances, non-collision metrics, render-based semantic consistency, and vision-language model verification. The result is a practical framework for turning text-level interaction intent into a geometrically grounded 3D human placement without task-specific training.

Keywords: human-scene interaction, foundation models, zero-shot synthesis, SMPL-X, contact optimization

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Thank advisors, supervisors, collaborators, funding sources, friends, and family here. Keep this section personal and direct.

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List of Abbreviations

HOI	Human-Object Interaction
ML	Machine Learning
RGB-D	Red, Green, Blue, and Depth

Chapter 1

Introduction

This chapter motivates zero-shot static 3D human-scene interaction synthesis from language, a calibrated RGB image, and a reconstructed scene mesh. It places the thesis in relation to learned HSI models, scene-constrained body optimization, and recent foundation-model-based systems, then states the problem formulation, approach, and contributions.

1.1 Why Human-Scene Interaction Matters

Indoor scenes are built around human action. Chairs are placed so that people can sit, tables so that people can work or eat, cabinets so that objects can be stored and retrieved, and handles so that drawers, doors, and appliances can be opened. The meaning of these objects is therefore not exhausted by their geometry or category labels. A chair is not only a shape occupying space; it affords support for a seated body. A drawer is not only a box-like structure inside a cabinet; it has a front face and a handle that invite a particular hand motion. A room becomes meaningful because human bodies can stand, reach, grasp, lean, sit, lie, and move through it.

This distinction is central to 3D scene understanding. Modern reconstruction methods can produce increasingly accurate models of indoor environments, including camera poses, dense geometry, object surfaces, and scene layout. However, a reconstructed scene by itself remains a passive description of the environment. It tells us where surfaces are, but not which surfaces can support the body. It shows object boundaries, but not which regions are functional for a task. It captures the state of a room, but not the ways in which a person could physically use that room. For applications that require human-centered reasoning, this missing layer is fundamental.

Human-scene interaction provides this missing layer by connecting scene structure to bodily action. Instead of asking only what objects are present, it asks how a person can engage with them. Where should the feet be placed when reaching for a cabinet? Which part of a sofa should support the pelvis and back during sitting? Where should the hand make contact when opening a drawer? What body orientation is plausible when lifting a trash bin? These questions require reasoning jointly about object function, human body parts, contact, balance, reachability, and spatial constraints. They cannot be answered by object recognition or scene reconstruction alone.

This capability is important for a broad class of downstream systems. In embodied AI and robotics, an agent must reason not only about object locations, but also about how humans act around those objects and which parts of the scene are functionally relevant. In augmented and virtual reality, inserted humans must appear grounded in the environment rather than floating near objects or intersecting with them. In animation, synthetic data generation, and digital twins, realistic human activity requires bodies that are placed according to the affordances of the scene. In human-centered scene analysis, the presence or absence of plausible interactions can reveal whether a reconstructed environment is usable, accessible, or semantically coherent.

The challenge is therefore not simply to add a human body to a 3D scene. A visually plausible body can still be physically or functionally wrong: a seated person may hover above a chair, a hand may reach the side of a cabinet instead of the drawer handle, or a body may intersect with furniture while appearing reasonable from one camera view. Human-scene interaction requires a stronger form of consistency. The body must be semantically aligned with the requested action, geometrically aligned with the scene, and physically plausible with respect to contact and collision.

This thesis studies human-scene interaction synthesis as a step toward scene representations that are not only geometric, but actionable. The goal is to synthesize human bodies that express how a real indoor scene can be used. By grounding human pose in scene affordances, object function, and physical contact, HSI synthesis moves beyond static reconstruction toward a more complete understanding of indoor environments as spaces for human activity.

1.2 From Reconstructed Geometry to Interaction Synthesis

A 3D reconstruction converts visual observations of a room into a metric representation of space. It provides camera poses, scene geometry, object surfaces, and the spatial layout of the environment. This representation is essential for grounding perception in real-world coordinates: it tells us where the floor is, how far a chair is from a table, where a cabinet is located, and which surfaces are visible or occluded. However, reconstruction alone does not specify how a human body should be arranged in the scene. Geometry describes the shape of the environment, but interaction requires reasoning about the relationship between that geometry and the body.

This gap becomes apparent even for simple activities. To synthesize a person sitting on a chair, it is not enough to know that a chair exists. The body must be placed so that the pelvis and thighs are supported by the seat, the feet remain near the floor, and the torso has a plausible orientation relative to the chair. To synthesize a person opening a drawer, it is not enough to locate the cabinet. The hand should reach the drawer pull or handle, while the body should stand at a reachable distance and face the object in a plausible way. To synthesize a person lifting a trash bin, the hands, arms, torso, and feet must form a coherent configuration around the object. In each case, the interaction is defined not only by the object category, but by the spatial and functional relation between body parts and scene parts.

Human-scene interaction synthesis addresses this problem by generating a human body configuration that is jointly consistent with an action and a 3D scene. Unlike image-based human insertion, the goal is not only to make a rendered view appear plausible. The synthesized body must exist in the same coordinate frame as the scene and must satisfy constraints that are meaningful in 3D: contact should occur at appropriate locations, the body should have a plausible scale and orientation, and severe interpenetration with the environment should be avoided. The output is therefore a scene-grounded human state, not simply a visual overlay.

This distinction separates interaction synthesis from several related tasks. Object detection identifies what is present in the scene, but it does not determine how the object can be used. Human pose estimation recovers a body configuration from observations, but it usually assumes that the person is already present. Text-to-image generation can produce visually convincing depictions of people interacting with objects, but the result is not necessarily metrically consistent with a reconstructed scene. Human-scene interaction synthesis combines aspects of all three problems while adding a stronger requirement: the generated body must be semantically aligned with the requested action and physically grounded in the 3D environment.

The difficulty is that interaction is structured. A single instruction often implies a set of relations among the action, the object, the relevant object part, the participating body parts, and the supporting scene surfaces. Sitting involves support relations between the lower body and a seat. Opening involves a reaching or grasping relation between a hand and a functional element. Lifting involves coordinated contacts between the hands and the object, together with a stable stance on the floor. These relations are not directly given by the reconstruction. They must be inferred from the task and then realized as a plausible body placement in the scene.

Interaction synthesis can therefore be viewed as the step from a passive scene model to an actionable scene model. A passive model answers geometric questions such as where surfaces are and how objects are arranged. An actionable model supports questions about what a human can do in the environment, where the body should be placed, which body parts should contact which scene regions, and whether the resulting configuration is physically credible. This thesis focuses on this transition: synthesizing a human body that does not merely appear inside a reconstructed room, but uses the room in a way that is consistent with language, geometry, and physical contact.

1.3 The Zero-Shot Challenge

Human-scene interaction synthesis becomes especially difficult when the system is expected to operate in a zero-shot setting. In a closed setting, the model may assume a fixed set of object categories, a fixed set of actions, or training examples that closely resemble the test scenes. Under these assumptions, the problem can be approached as learning regularities from paired examples of humans interacting with objects. The model can learn, for instance, that people sit on chairs, stand on floors, touch tables with their hands, or lie on beds. However, such regularities are only reliable when the test interaction lies within the distribution of the training data.

Real indoor scenes do not satisfy this assumption. They contain diverse object instances, cluttered layouts, partial visibility, unusual viewpoints, and functional elements that vary widely in appearance. A drawer handle may be horizontal, vertical, recessed, metallic, wooden, large, small, or partially occluded. A chair may be an office chair, dining chair, stool, sofa, or bench. Even when the object category is familiar, the region relevant to an interaction may differ across instances. The same action phrase can therefore require different geometric solutions depending on the scene. 'Open the drawer" requires the hand to reach the handle of a particular drawer in a particular cabinet. 'Sit on the chair" requires the body to find a support surface of the correct height and orientation. "Lift the bin" requires a graspable region on the visible object and a stable full-body configuration around it.

This variability exposes the limitation of relying only on supervised 3D human-scene interaction data. Paired HSI data is expensive to collect because it requires reconstructed scenes, accurate human body fits, object annotations, action labels, contact information, and enough diversity in body shapes, object instances, and room layouts. As a result, existing datasets cover only a small subset of the possible interactions that occur in indoor environments. Learned HSI models can produce strong results for common interactions represented in their training sets, but they often depend on closed vocabularies, predefined object categories, known target objects, or interaction patterns observed during training. These constraints make it difficult to synthesize interactions for arbitrary language instructions in previously unseen reconstructed scenes.

Recent work has therefore explored the use of large vision-language, image-generation, and video-generation models as sources of interaction knowledge. These models have seen broad visual and linguistic evidence of people using objects, and they can often suggest plausible body poses, object affordances, and interaction layouts without being trained directly on 3D HSI datasets. This has motivated zero-shot and training-free approaches to human-scene interaction synthesis, where foundation models provide semantic or visual priors and 3D geometry provides metric grounding. Methods such as GenZI, FunHSI, and ZeroHSI show that this direction is promising because interaction knowledge can be transferred from large-scale visual models to 3D scenes without requiring paired 3D interaction supervision.

However, zero-shot HSI is not solved by generating a plausible image of a person interacting with an object. A generated image may express the intended action visually, but it does not guarantee that the body is correctly placed in the reconstructed scene. The hands may appear near an object in image space while missing the corresponding 3D surface. The body may look plausible from one camera view while penetrating furniture or floating above the floor. The generated interaction may also contact the wrong part of the object: the side of a cabinet instead of the drawer handle, the rim of a bin instead of a graspable side, or the back of a chair instead of the seat. The challenge is therefore to convert broad interaction priors into scene-specific, metrically valid human-body configurations.

The zero-shot setting thus requires solving several coupled questions. Given an open-vocabulary instruction, the system must infer which scene object is relevant, which part of that object affords the action, which human body parts should participate, where supporting contacts should occur, and how the full body should be arranged in 3D. These questions cannot be answered independently. The choice of contact region affects the pose; the pose affects the feasible body location; the body location affects reachability, balance, and collision; and the scene geometry constrains all of these decisions. Zero-shot HSI is therefore not simply a problem of text-conditioned pose generation, but one of grounding an inferred interaction structure in a specific reconstructed environment.

This thesis addresses the zero-shot challenge in the static 3D setting. Rather than assuming a fixed action vocabulary or paired 3D interaction examples for every task, it treats the instruction as an open-ended description of how a person should use the scene. The goal is to infer the relevant interaction structure and synthesize a single SMPL-X body that realizes that structure in the original scene coordinate frame. This formulation keeps the problem narrower than full dynamic interaction generation, but it preserves the central difficulty of zero-shot HSI: the system must transform language and scene observations into a physically plausible human-scene contact configuration without relying on task-specific 3D supervision.

1.4 Body-Part–Scene-Part Grounding

The previous sections framed human-scene interaction synthesis as more than placing a plausible body near a relevant object. The remaining question is what structure should connect the language instruction, the human body, and the scene. A natural first answer is to reason at the object level: a person sits on a chair, opens a drawer, touches a table, or lifts a bin. Object-level reasoning is useful, but it is not precise enough for interaction synthesis. Most interactions are not defined by whole objects. They are defined by relations between specific parts of the human body and specific regions of the scene.

Consider the instruction 'a person opening a drawer." The target object may be a cabinet or drawer unit, but the interaction itself is concentrated at the drawer handle or front edge. The relevant human part is usually a hand, not the entire body. At the same time, the rest of the body is not arbitrary: the feet must remain supported by the floor, the torso must be oriented toward the cabinet, and the arm must be arranged so that the hand can reach the functional part. Similarly, 'a person sitting on a chair" primarily requires support between the lower body and the seat, but also implies a plausible relation between the feet and the floor and between the torso and the chair orientation. These examples show that an interaction is best understood as a set of body-part–scene-part relations.

This thesis adopts that view. Instead of treating the human body as a single object and the scene as a collection of object categories, it represents an interaction through participating body regions, scene regions, and contact relations between them. The human body is decomposed into interaction-relevant parts, such as hands, arms, feet, pelvis, thighs, back, and torso. These parts correspond to regions on the SMPL-X surface that commonly participate in physical interactions. On the scene side, the relevant regions may be broad support surfaces, such as floors, seats, beds, and tabletops, or smaller functional elements, such as handles, knobs, switches, faucets, and drawer pulls. The task of HSI synthesis is then to arrange the body so that the appropriate body regions are grounded to the appropriate scene regions.

This formulation is important because different actions require different kinds of contact. Support interactions, such as sitting, lying, standing, or leaning, depend on stable contact between load-bearing body regions and supporting scene surfaces. Manipulation interactions, such as opening, pulling, pressing, or lifting, depend on localized contact between the hands and functional object parts. Many interactions combine both types. A person opening a drawer must stand on the floor while reaching the handle. A person lifting a bin must grasp the object while maintaining a stable stance. A person leaning over a sink may contact the floor with the feet, the counter with the hands or torso, and the faucet with one hand. Representing these interactions as structured body-scene contacts makes these dependencies explicit.

The body-side representation also matters. SMPL-X provides a detailed parametric human mesh, but the mesh vertices themselves are not semantically meaningful unless they are grouped into interaction regions. This thesis therefore uses manually defined contact regions on the SMPL-X body surface corresponding to interaction parts such as hands, arms, feet, pelvis, back, and other task-relevant regions. These regions provide a bridge between high-level language and low-level geometry: an instruction may mention a hand, a seated posture, a leaning action, or a grasping motion, while the final synthesis must place specific surface regions of the body near specific surface regions of the scene.

The scene-side representation is equally important. For many tasks, detecting the object category or even the object part is insufficient. The same cabinet can support several different interactions: opening a drawer, reaching for an upper shelf, leaning on the counter, or standing beside it. Each task selects a different scene region and a different set of body regions. Conversely, the same body part can participate in different functional roles: a hand can grasp a handle, press a switch, support the body on a tabletop, or hold the side of a bin. Interaction synthesis must therefore resolve not only which object is involved, but which object region is relevant and what type of body contact it requires.

Moreover, part-level grounding alone does not fully determine the interaction. Many object parts are spatially extended, and the exact location of contact within a detected part can strongly affect the plausibility of the synthesized pose. For example, when lifting a table, it is not enough to identify the tabletop as the relevant part; the hands should contact plausible grasp or support locations on the tabletop, often near accessible edges or corners rather than at arbitrary points on the surface. Similarly, for an interaction involving a ladder, the system must reason not only about the ladder as an object, or about its rails and rungs as parts, but also about which specific rail and rung regions should be contacted by the hands and feet. The same rung may support a foot at one location while a hand grasps a nearby rail at another, and these local choices determine whether the resulting pose is physically and functionally coherent.

This distinction is important because semantic part labels provide structure, but they do not by themselves specify contact geometry. A tabletop, rail, rung, seat, counter, cabinet front, or floor region can contain many possible contact locations, only some of which are compatible with the requested action, the human pose, the object affordance, and the surrounding scene. Body-part–scene-part grounding therefore requires a finer notion of localization: the framework must identify not only the relevant object and object part, but also the appropriate contact region within that part. This finer grounding allows the synthesized body to interact with the scene in a way that is both semantically correct and geometrically plausible.

To capture this structure, the thesis uses a Scene Interaction Graph as an intermediate representation. The graph expresses the target scene element, the participating human interaction parts, and the expected contact or support relations between them. Its purpose is not to describe every detail of the final pose, but to specify the interaction constraints that the pose should satisfy. In this sense, the graph acts as a compact bridge between open-language task descriptions and 3D body synthesis. It turns an instruction such as "lift the trash bin with both hands" into a structured interaction hypothesis involving the bin, the two hands, the arms, the feet, and the supporting floor.

A key aspect of this thesis is that the scene-side contact regions are not assumed to be given by manual annotations, object geometry alone, or part labels alone. Instead, they are inferred through image-based generative reasoning. Starting from the scene image, a diffusion-based inpainting model first synthesizes a plausible human performing the requested interaction in the scene. This generated human provides a visual hypothesis about pose, orientation, laterality, and approximate contact layout. Then, conditioned on the interaction parts specified by the Scene Interaction Graph, a second diffusion-based inpainting step marks the corresponding contact regions on the target object or support surface in the original scene image. In this way, the scene image is used twice: first to imagine how the interaction should look, and then to localize where the relevant body parts should make contact on the object or scene.

This image-based contact grounding is particularly valuable for functional interactions, where the relevant contact region may be small, localized, or ambiguous even when the correct object part is known. A cabinet may contain many possible surfaces, but only the handle or drawer pull is appropriate for opening. A trash bin may have several visible faces, but only some are plausible grasp regions for lifting. Likewise, a large tabletop, ladder rail, or ladder rung may be semantically correct at the part level, but the interaction still depends on where within that part the hands, feet, or other body regions should make contact. By using diffusion models to propose both the human interaction hypothesis and the object-side contact regions, the framework leverages broad visual priors while keeping the final interaction tied to the actual scene and to the fine-grained spatial layout of the interaction.

Body-part–scene-part grounding is therefore the central abstraction behind this thesis. It makes the interaction explicit enough to guide synthesis, but flexible enough to support open-vocabulary instructions and previously unseen scenes. By grounding actions in relations between human surface regions and scene surface regions, and by further localizing contact within those scene regions, the formulation moves beyond object-level and part-level semantics toward a representation of how the body physically and functionally engages with the environment.

1.5 Problem Setting and Contributions

This thesis studies zero-shot static 3D human-scene interaction synthesis in real reconstructed indoor scenes. The input consists of a natural-language instruction, a calibrated

RGB image of the scene, and a reconstructed scene mesh. The output is a single SMPL-X body placed in the original scene coordinate frame. The synthesized body should express the requested action, contact the appropriate scene regions with the appropriate body parts, and remain physically plausible with respect to scale, orientation, support, and collision.

The setting is zero-shot because the method is not trained on paired 3D examples of the requested interaction in the target scene. The instruction may describe common support interactions, such as sitting on a chair, lying on a bed, or leaning on a table, or functional interactions, such as opening a drawer, pressing a switch, turning a faucet, or lifting a bin. In each case, the system must infer the relevant scene element, the functional or supporting region on that element, the participating human body parts, and the spatial arrangement required to realize the interaction. The target object and the required contact regions are not assumed to be available as ground-truth annotations.

The formulation is static. The goal is not to synthesize a full motion sequence, simulate forces, or modify the state of the scene. Instead, the output represents a single interaction state. For steady actions such as sitting or lying, this state corresponds to the interaction itself. For actions involving movable or articulated objects, such as opening a drawer, pulling a door, or lifting a trash bin, the synthesized body represents the pre-action or contact-ready state: the human has reached the relevant scene region and adopted a plausible pose, but the object has not yet moved. This choice isolates a fundamental subproblem of human-scene interaction: determining where contact should occur and how the body should be configured in a real 3D environment.

The proposed approach is built around body-part–scene-part grounding. An open-language instruction is first interpreted as a structured interaction involving human interaction regions, scene regions, and contact relations. On the human side, SMPL-X is treated not only as a parametric body model, but as a surface with semantically meaningful interaction regions, such as hands, arms, feet, pelvis, thighs, back, and torso. On the scene side, the relevant regions may be broad support surfaces or localized functional elements. The synthesis problem is then to place the body so that these body regions and scene regions are brought into physically and semantically meaningful correspondence.

To support this grounding in open-vocabulary scenes, the thesis combines structured interaction reasoning with generative visual priors and scene-aware 3D fitting. A Scene Interaction Graph represents the expected relations among the action, target scene element, participating body parts, and contact regions. Diffusion-based image reasoning provides visual hypotheses about how the interaction should appear and where the corresponding object-side contacts should lie. These hypotheses are then grounded in the reconstructed scene to synthesize a metric SMPL-X body that satisfies the inferred interaction structure. At a high level, the method uses foundation models to reason about what interaction is plausible, while using the scene reconstruction to ensure that the final body is placed in

3D rather than merely depicted in an image.

This thesis makes the following contributions:

1. **A zero-shot formulation of static 3D human-scene interaction synthesis.** The thesis defines the task of synthesizing a single SMPL-X body in a real reconstructed indoor scene from an open-language instruction, a calibrated RGB image, and a scene mesh, without requiring paired 3D interaction examples for the target task.

2. **A body-part–scene-part grounding view of HSI.** The thesis formulates interaction synthesis as the problem of grounding relations between human interaction regions and scene interaction regions, rather than only matching whole-body poses to object categories.

3. **A Scene Interaction Graph representation for open-vocabulary interactions.** The thesis introduces a structured representation that captures the target scene element, participating human body parts, scene regions, and expected contact or support relations implied by a natural-language instruction.

4. **A SMPL-X interaction-part abstraction for contact-aware synthesis.** The thesis defines semantically meaningful contact regions on the SMPL-X body surface, enabling high-level interaction descriptions involving hands, arms, feet, pelvis, back, and other body regions to be connected to low-level 3D body geometry.

5. **An image-guided strategy for localizing scene-side contact regions.** The thesis uses generative image reasoning to infer where the human interaction parts should contact the target object or support surface, allowing functional regions such as handles, knobs, drawer pulls, and graspable object areas to be identified without manual scene-side contact annotation.

6. **A scene-grounded synthesis framework for physically plausible body placement.** The thesis synthesizes a full SMPL-X body in the reconstructed scene coordinate frame, using the inferred interaction structure and scene geometry to encourage semantic consistency, appropriate contact, realistic scale, and reduced physical artifacts such as floating or penetration.

Together, these contributions move static HSI synthesis beyond object-level placement and toward interaction-level grounding. The thesis argues that plausible human-scene interaction requires knowing not only what object is involved, but which parts of the human body should engage which parts of the scene, and how that engagement can be realized as a coherent 3D body configuration.

1.6 Thesis Structure

Chapter 2 reviews the foundations and related work necessary for this thesis, including 3D scene reconstruction, parametric human body models, human-scene interaction synthesis, affordance and contact reasoning, foundation models for visual generation, and evaluation of physically grounded human-scene configurations.

Chapter 3 presents the proposed zero-shot static HSI synthesis framework. It describes the Scene Interaction Graph, the SMPL-X interaction-part abstraction, diffusion-based human and contact-region reasoning, scene-side contact grounding, and the final scene-aware SMPL-X fitting procedure.

Chapter 4 defines the experimental setup, including the ScanNet++ scenes, interaction instructions, selected action categories, implementation details, baselines or ablations, and quantitative and qualitative evaluation metrics.

Chapter 5 reports the experimental results and analyzes the behavior of the method. It examines semantic consistency, contact accuracy, physical plausibility, pose diversity, qualitative examples, ablation studies, and common failure modes.

Chapter 6 summarizes the findings of the thesis and discusses limitations and future directions, including dynamic interaction synthesis, richer physical reasoning, multi-view grounding, multi-human interactions, and more complete modeling of object state changes.

Chapter 2

Related Work

This chapter positions the thesis within prior work on human-scene interaction synthesis. It organizes the literature along two axes that are central to the thesis: trained versus zero-shot methods, and static 3D versus dynamic 4D interaction synthesis.

2.1 Contact Representations and Supervision for 3D HSI

Human-scene interaction requires more than placing a plausible human pose inside a 3D environment. A synthesized body must also be geometrically compatible with the surrounding scene: it should be supported by appropriate surfaces, avoid severe interpenetration, and make contact with scene regions that are meaningful for the intended interaction. This distinction is especially important for static 3D HSI, where the output is a single body configuration rather than a motion sequence. In such a setting, contact provides one of the main cues that separates a merely plausible pose from a physically and functionally grounded interaction.

The scene representation determines whether such contact can be enforced metrically. Indoor reconstruction datasets such as ScanNet and ScanNet++ provide RGB observations, camera calibration, and reconstructed scene geometry, allowing image-space evidence to be related to a common 3D coordinate frame [1, 2]. This calibration is essential for methods that reason jointly in image and scene space. A contact cue observed in an RGB view is only useful for 3D synthesis if it can be associated with a surface in the reconstructed scene. In this thesis, ScanNet++ provides the calibrated image, camera parameters, and scene mesh on which body-scene contact constraints are ultimately enforced.

On the human side, parametric body models provide a differentiable representation for optimization. SMPL-X is particularly suitable for human-scene interaction because it represents the full body as a posed mesh with articulated body, hand, face, shape, and global transformation parameters [3]. Its dense surface makes it possible to define contact not only at sparse joints, but over anatomically meaningful body regions. For example, grasping, standing, leaning, sitting, and pushing involve different parts of the body surface. Therefore, contact-aware HSI methods often require a mapping from semantic body parts to subsets of body vertices, rather than treating the body as an undifferentiated mesh.

Scene-aware human fitting demonstrated early that image-based human estimates are often physically invalid when evaluated in 3D. PROX extended monocular SMPL-X fitting with constraints from reconstructed scenes, using contact and interpenetration terms to reduce ambiguities in human pose estimation [4]. Its central insight is directly relevant to synthesis: a body may explain the image evidence while still floating above the floor, penetrating furniture, or failing to touch a supporting surface. By introducing the scene as an active constraint, PROX established contact and collision avoidance as fundamental components of 3D human-scene reasoning.

Subsequent work developed contact into a learned representation for HSI. PLACE models proximity and contact between an articulated human body and a 3D scene, emphasizing that realistic bodies in scenes depend on local body-scene relationships rather than only global placement [5]. POSA further represents interaction in a body-centric way by augmenting SMPL-X vertices with contact probabilities and semantic scene labels [6]. These methods show that contact can serve as an intermediate representation between human pose and scene affordance. They also motivate body-surface reasoning: plausible HSI depends not only on whether a person is near an object, but on which body surface is near or touching which scene surface.

Dense contact datasets and predictors further highlight the importance of body-level contact supervision. RICH introduced high-quality human-scene capture with dense vertex-level body-scene contact labels, and BSTRO uses this data to infer body-scene contact from a single image [7]. DECO extends dense 3D contact estimation to in-the-wild images by predicting vertex-level contact on the human body using both body-part and scene-context cues [8]. These works are important because they move beyond coarse contact categories and sparse joints. They treat contact as a dense surface property, which is more appropriate for physical interaction than global action labels alone.

Human-object interaction datasets provide another source of contact supervision. GRAB captures whole-body grasping motions with detailed body, hand, object, and contact information [9]. BEHAVE provides multi-view RGB-D recordings with 3D human and object fits, together with contact annotations for full-body human-object interaction [10]. These datasets are valuable because they capture how the full body participates in object interaction, not only the hands. However, they are primarily object-centric. The target object is usually explicitly present and tracked, and the interaction is observed or captured rather than inferred from an open-vocabulary instruction in an arbitrary reconstructed room.

Together, these works establish that contact is a central representation for 3D HSI. They also reveal several limitations that matter for this thesis. First, many methods model contact primarily on the human side, for example by predicting which SMPL or SMPL-X vertices are in contact. This is useful for understanding the body, but it does not by itself identify the exact scene surface that should be contacted in a new reconstructed environment. Second, contact supervision is often tied to specific datasets, object categories, scenes, or captured interaction types. This limits generalization to open-vocabulary instructions and unseen scene layouts. Third, object-level grounding is often too coarse. Even when the correct object is known, the method must still determine the appropriate contact patch: a hand lifting a table should contact a plausible region of the tabletop or edge, a hand opening a washing machine should contact the door or handle region, and a foot climbing a ladder should contact a rung or step rather than the object instance as a whole.

This thesis builds on the contact-centered view of HSI, but uses contact differently. Instead of learning a contact prior from paired human-scene interaction data, the method defines body-side contact regions on SMPL-X and estimates the corresponding scene-side contact regions from visual interaction evidence. The scene-side target may be a functional part, such as a handle or control, but it may also be a pose-dependent contact patch on a larger surface, such as a tabletop, sofa seat, wall, rail, or floor. The resulting image-space masks are projected into the reconstructed scene mesh and used as explicit 3D contact targets for optimization. In this way, the method inherits the geometric accountability of contact-aware HSI while addressing a broader problem: open-vocabulary, body-part-specific scene-side contact grounding in calibrated reconstructed scenes.

2.2 Static and Language-Controlled HSI Synthesis

While contact representations define how human-scene compatibility can be described, HSI synthesis methods address the generative problem: producing a human body or motion that is plausible in a given scene. In the static 3D setting, the goal is to synthesize a single body configuration that is semantically meaningful and physically feasible with respect to the environment. This requires deciding where the body should be placed, how it should be oriented, which pose it should take, and how it should relate to nearby scene surfaces. Compared with scene-agnostic pose generation, static HSI synthesis must account for both the semantic role of the scene and its metric geometry.

Early work on 3D human affordance prediction framed this problem as estimating which human poses are supported by an indoor environment. Putting Humans in a Scene predicts human affordances in 3D indoor scenes by learning what poses are plausible for a given scene, such as sitting on a chair or standing on the floor [11]. The method combines semantic knowledge from 2D human poses with geometric knowledge from 3D indoor scenes, allowing it to predict semantically plausible and physically feasible human configurations. This line of work is important because it treats the scene as an active source of affordances rather than as a passive background. However, the interaction is still controlled indirectly through learned affordance distributions, not through open-ended language instructions or body-part-specific contact requirements.

Generating 3D People in Scenes without People further develops static HSI synthesis by generating full SMPL-X bodies in empty 3D scenes [12]. The method first predicts semantically plausible body configurations conditioned on a latent scene representation and then refines the generated bodies with scene constraints to improve physical feasibility. This two-stage structure is influential: a learned model proposes a plausible human-scene configuration, while a refinement stage enforces physical constraints such as contact support and non-penetration. The method demonstrates that realistic static HSI requires both semantic compatibility and geometric correction. Nevertheless, the generated interactions are primarily governed by learned scene affordances rather than by a user-specified, open-vocabulary interaction with explicit contact intent.

Contact-aware synthesis methods such as PLACE and POSA also contribute to static HSI generation. PLACE uses local scene geometry to synthesize articulated humans that are compatible with nearby surfaces, emphasizing proximity and contact between the body and the environment [5]. POSA learns a body-centric representation of contact and scene semantics on SMPL-X, allowing plausible body placements to be generated or refined according to predicted body-scene relationships [6]. These methods show that contact is not only useful for evaluating a generated body, but can also guide synthesis itself. However, because the contact priors are learned from available interaction data, their behavior is tied to the distribution of scenes, poses, and contact configurations seen during training.

Semantic control is introduced more explicitly in COINS, which synthesizes humans interacting with 3D scenes from action-object specifications [13]. Instead of generating a plausible body anywhere in the scene, COINS allows the user to specify interactions such as sitting on a chair or touching a table. This is an important step from generic scene population toward controllable HSI synthesis. COINS also supports compositional interaction specifications, making it possible to combine simpler action-object relations into more complex human-scene configurations. However, the control signal remains a predefined action-object pair. This is suitable for interactions whose geometry is largely determined by the object category and action label, but it is less expressive for instructions that require fine-grained contact reasoning, such as selecting where on a large table surface the hands should touch, or which visible part of an object should be contacted.

The distinction between object-level control and contact-region control is important for this thesis. A semantic specification such as ‘touch table” or ‘sit chair” identifies the target object and broad action, but it does not necessarily determine the exact scene-side contact patch. Even when the relevant object part is large, such as a tabletop, sofa seat, wall, or floor, the precise contact region should still be consistent with the intended pose, body scale, orientation, and scene layout. For functional interactions, the required contact may be even more localized, such as a handle, rail, rung, button, or edge. Therefore, static HSI synthesis requires more than choosing a plausible object and body pose; it may also require grounding each relevant body part to a specific region of the scene surface.

Language-conditioned HSI extends semantic control beyond fixed action-object labels, but most existing work in this direction focuses on motion rather than a single static body configuration. HUMANISE introduced language-conditioned human motion generation in 3D scenes by aligning motion sequences with indoor scenes and language descriptions [14]. This formulation shows that natural language can specify not only the action but also the interacting object and spatial intent. TextSceneMotion further decomposes scene-aware text-to-motion generation into target-object grounding and object-centric motion generation [15]. This decomposition is relevant because it separates the problem of understanding the language target from the problem of generating the human behavior around that target. TeSMo similarly studies text-controlled scene-aware motion generation using diffusion models, producing motions such as navigation and object interaction in 3D scenes [16].

Other dynamic HSI methods broaden the scope from static placement to long-horizon scene-aware behavior. SceneDiffuser formulates scene-conditioned generation, optimization, and planning as a diffusion process applicable to tasks such as human pose generation, human motion generation, grasping, and navigation [17]. TRUMANS scales dynamic HSI supervision by capturing diverse human interactions in indoor scenes and training a model for scene-conditioned interaction sequences [18]. These works are important because they show how language, motion priors, scene context, and contact can be integrated for temporally evolving interactions. However, their central problem is different from the one considered in this thesis. They must model trajectories, transitions, temporal coherence, and changing contacts over time, whereas the present work focuses on one static interaction state.

Overall, static and language-controlled HSI synthesis has progressed from affordance-based human placement, to contact-aware body generation, to semantic and language-conditioned interaction control. These methods establish that plausible HSI depends on both scene semantics and physical compatibility. However, most data-driven approaches remain constrained by learned interaction distributions, predefined action-object labels, or paired scene-motion supervision. They also typically ground interaction at the level of an object, action, or motion trajectory, rather than explicitly estimating the body-part-specific scene regions that should participate in contact. This leaves two coupled challenges for zero-shot static 3D HSI: first, extracting interaction-specific 2D contact regions on the scene from visual and language evidence; and second, converting those regions into metric 3D contact targets that can guide body optimization. The next section discusses foundation-model-based methods that address parts of this problem, and motivates the need for an explicit contact-localization and geometric-grounding stage.

2.3 Foundation-Model and Zero-Shot Interaction Generation

The methods discussed so far rely primarily on learned interaction priors, paired human-scene data, predefined action-object labels, or motion supervision. Recent foundation-model-based approaches follow a different strategy: instead of training a task-specific HSI model from paired 3D interaction data, they use pretrained language, vision-language, image generation, video generation, segmentation, and human reconstruction models as sources of interaction knowledge. These models have absorbed broad visual and semantic regularities about people, objects, actions, and everyday scenes. As a result, they can provide useful priors for interactions that are difficult to cover exhaustively with captured 3D HSI datasets.

In zero-shot HSI, foundation models are typically not used as final geometric solvers. A language model may infer the likely interaction structure from a text prompt, a vision-language model may judge whether a generated image satisfies an instruction, and an image generation model may synthesize a plausible human interacting with the scene. However, these outputs are usually expressed in language or image space. They do not directly provide a body mesh in the scene coordinate frame, nor do they automatically identify which reconstructed scene vertices should be contacted by which body parts. Therefore, zero-shot HSI requires an additional grounding step that converts foundation-model hypotheses into metric constraints.

GenZI is a key example of zero-shot static 3D HSI [19]. Given a 3D scene, an interaction description, and a specified interaction location, GenZI uses large vision-language and image inpainting models to generate plausible 2D human-scene interaction hypotheses from rendered scene views. These generated images provide visual evidence about the human pose and the intended interaction, after which a 3D human body is optimized in the scene. GenZI demonstrates that pretrained 2D models contain useful priors for 3D human-scene interaction, even without training on paired 3D HSI data. At the same time, the generated image mainly acts as an implicit interaction hypothesis. The method does not explicitly extract body-part-specific scene-side contact masks and use them as direct metric targets for optimization.

FunHSI is more directly concerned with open-vocabulary functional interaction [20]. It observes that many interaction prompts cannot be solved by object category alone: a person increasing room temperature, opening a window, or using a particular appliance must reason about the functional elements of the scene and the body configuration required to interact with them. FunHSI uses foundation models to identify functional scene elements, construct contact graphs, synthesize a human performing the task, and refine the 3D body through optimization for physical and functional correctness. This is closely related to the motivation of this thesis because both approaches treat open-vocabulary interaction as a contact-grounding problem rather than merely as object-level placement.

The present thesis differs in the way the scene-side contact is grounded. Instead of relying primarily on explicit functional-element identification, the proposed method first represents the interaction as body-part-to-scene contact requirements through a Scene Interaction Graph. The scene node may be a target object or the floor, and the precise contact patch is inferred later from visual evidence. This formulation covers functional contacts, such as touching a handle or control, but it also covers broader pose-dependent contacts, such as choosing where on a tabletop, wall, sofa seat, treadmill rail, or floor the body should touch. Thus, the problem is not only functional object-part discovery. It is broader: interaction-specific scene-side contact localization for each required body part.

This distinction is important because the difficult step is not only projecting 2D contact evidence into 3D. Projection is necessary, but it assumes that an appropriate image-space contact region has already been identified. In many interactions, obtaining that 2D region is itself ambiguous. A generated person may visually suggest that a hand is lifting a table, but the method must still decide which visible patch of the tabletop or edge corresponds to the hand contact. A person may be shown reaching toward a washing machine, but the correct hand contact should lie near the door or handle region rather than an arbitrary visible surface. A broad object mask is not sufficient in either case. The contact region must be consistent with the instruction, the generated body pose, the body-part identity, the visible object geometry, and the original calibrated scene image.

A related development outside HSI is the use of foundation image models as components of agentic 3D reasoning systems. WorldAgents investigates whether 2D foundation image models can be used as agents for 3D world synthesis, using a multi-agent architecture with a vision-language director, an image generator, and a verifier that evaluates generated frames in both image and reconstruction space [21]. Although its goal is 3D world generation rather than human-scene interaction, it supports a broader observation that is relevant here: foundation image models become more useful for 3D tasks when embedded in iterative generate-and-verify loops rather than used as one-shot image synthesizers.

This motivates the agentic contact-localization stage used in this thesis. The generated human frame is treated as a visual reference for pose, orientation, laterality, and likely contact layout, while the original scene image remains the authoritative canvas. An image model proposes colored overlays for the required scene-side contact regions; the colored pixels are extracted and recomposited onto the unedited original image; and a vision-language evaluator checks whether the resulting composite correctly marks the contact regions implied by the generated interaction. If the contact masks are incorrect, a correction instruction is issued and the process is repeated. The output of this loop is not a final image, but a set of verified body-part-specific 2D contact regions that can be projected to the reconstructed scene mesh.

Several zero-shot human-object interaction methods are related to this direction. DreamHOI uses text-to-image diffusion priors for zero-shot 3D human-object interaction synthesis with a given human and object, optimizing the human articulation so that the body appears to interact realistically with the object [22]. InterFusion addresses text-driven 3D human-object interaction generation by using text-derived human pose priors and a two-stage generation process to reduce the difficulty of directly producing a full interaction scene from text [23]. InteractAnything uses large language models and diffusion-based object affordance parsing for zero-shot 3D human-object interaction, including extraction of contact points for open-set objects [24]. These works show that foundation models can support interaction synthesis beyond closed object and action datasets.

However, object-centric HOI differs from full-scene HSI. In HOI, the target object is usually explicit, isolated, or provided as an input asset. In HSI, the object is embedded in a larger scene with surrounding geometry, occlusions, support surfaces, and other objects. A body may need to contact both the target object and the floor, avoid nearby obstacles, and remain consistent with the original reconstructed environment. Therefore, scene-side contact grounding in HSI must account not only for object affordance, but also for camera calibration, visible scene geometry, mesh projection, and body-scene collision constraints.

Part-level affordance reasoning in HOI is nevertheless relevant. HOI-PAGE introduces Part Affordance Graphs for zero-shot human-object interaction generation, representing how human body parts engage with object parts and using these relations to guide 4D HOI synthesis [25]. This supports an important trend: realistic interaction generation increasingly requires structured relations between body parts and scene or object parts, rather than only global text prompts or whole-object proximity. The Scene Interaction Graph used in this thesis follows a related motivation, but it is intentionally simpler on the scene side. It specifies which body regions should contact the target object or floor, while the exact scene-side contact region is estimated visually from the generated interaction and original scene image.

Foundation-model priors have also been explored for dynamic HSI. ZeroHSI uses video generation models as motion priors for zero-shot 4D human-scene interaction, reconstructing temporally evolving human-scene interactions through neural human rendering and differentiable optimization [26]. GenHSI similarly uses pretrained video diffusion models for controllable human-scene interaction video generation, decomposing the process into script writing, 3D-aware keyframe previsualization, and animation [27]. These works show that generated videos can provide priors over human motion and interaction dynamics. Their goal, however, is temporal visual synthesis or 4D interaction generation, whereas this thesis focuses on a single static body configuration with explicit metric contact constraints.

Overall, zero-shot interaction generation methods show that foundation models provide useful semantic, visual, and motion priors for HSI and HOI. They reduce dependence on paired 3D interaction datasets and enable broader language control. The remaining challenge is to make these priors geometrically accountable. For static 3D HSI, this means first extracting interaction-specific 2D contact regions on the original scene image, and then projecting those regions to the reconstructed mesh as scene-side targets for body optimization. The proposed method addresses this gap by using foundation models not only to generate a plausible human interaction image, but also to localize and verify body-part-specific contact regions before enforcing them through SMPL-X optimization.

2.4 Research Gap and Positioning

Prior work has established several important ingredients for human-scene interaction synthesis. Contact-aware 3D HSI methods show that plausible bodies in scenes require contact and collision reasoning. Static synthesis methods show that humans can be generated in 3D environments using learned scene affordances, body-scene compatibility priors, or semantic action-object control. More recent zero-shot methods show that foundation models can provide useful visual and semantic priors for open-vocabulary interactions. However, these directions do not fully address the problem considered in this thesis: synthesizing a static 3D human in a reconstructed scene from an open-vocabulary instruction while grounding each required body contact to a specific scene-side region.

The key missing step is interaction-specific scene-side contact grounding. Object-level grounding is often too coarse, because knowing the target object does not determine where the body should touch it. Part-level grounding is also not always sufficient. Some interactions involve small functional parts, such as handles, controls, rungs, or buttons, but others involve large object parts or broad support surfaces. For example, lifting a table may require hand contact on a plausible region of the tabletop or table edge; sitting on a sofa requires selecting an appropriate patch on the seat or back support; and leaning on a wall requires choosing a contact region consistent with the body pose and orientation. Thus, the desired contact target is not simply an object instance, and sometimes not even an entire semantic object part, but a localized body-part-specific region of the visible scene surface.

Foundation-model-based methods help by producing plausible visual interaction hypotheses, but these hypotheses are not directly usable as 3D constraints. A generated image may suggest that a hand touches an object, but it does not automatically provide a reliable 2D mask for the contacted scene region, nor a corresponding set of 3D mesh vertices. Therefore, zero-shot static HSI requires two linked operations: first, extracting the correct 2D contact regions on the original scene image; and second, projecting those regions into the reconstructed scene as metric contact targets.

This thesis addresses this gap through an agentic contact-grounding pipeline. A Scene Interaction Graph first specifies which body regions should contact the target object or floor. A generated human frame then provides a visual hypothesis for pose, orientation, laterality, and approximate contact layout. Instead of accepting this generated image as the final output, the method uses it to guide contact localization on the original calibrated scene image. Colored contact overlays are generated, extracted, recomposited onto the unchanged original image, and verified by a vision-language evaluator. This closed-loop process produces body-part-specific 2D contact masks while reducing the effect of scene drift or incorrect one-shot annotations.

The accepted masks are then projected to the reconstructed scene mesh and used as explicit scene-side targets for SMPL-X optimization. The final body is initialized from monocular human reconstruction and refined using contact attraction, penetration avoidance, and initialization priors. This distinguishes the thesis from prior work in three ways: it does not require paired 3D HSI training data, it is not limited to predefined action-object labels, and it treats generated visual evidence as an intermediate source of contact constraints rather than as a final visual result. The contribution is therefore the conversion of open-vocabulary visual interaction evidence into metrically grounded body-part-specific contact targets for static 3D human-scene interaction synthesis.

Chapter 3

Methodology

This chapter describes the proposed zero-shot static human-scene interaction synthesis pipeline. The method converts a natural-language instruction, a posed ScanNet++ RGB view, and a reconstructed scene mesh into a single SMPL-X body placed in the scene coordinate frame.

3.1 Overview

This chapter presents a zero-shot pipeline for static 3D human–scene interaction synthesis. The method is zero-shot in the sense that it requires no interaction-specific training: it relies only on pretrained foundation models, together with explicit camera geometry and optimization. Its input is a natural-language instruction y, a single posed RGB view I of a ScanNet++ scene with known intrinsics K and extrinsics (R_c, t_c), and the reconstructed scene mesh M_s. Its output is a single SMPL-X body placed in the scene coordinate frame. The result is a *static interaction state*: one body configuration that realizes the requested interaction, rather than a motion sequence or an object-state change. For object-manipulation instructions, the object itself remains fixed in the reconstructed scene.

Throughout this chapter, a single running example illustrates each stage: the instruction "a person running on the treadmill, one foot landing on the moving belt, the other leg lifted behind, and both hands lightly holding the side handles." This example exercises the full method, since it requires two hand–object contacts, an asymmetric foot contact on a functional surface (the belt), and a lifted leg with no support.

Interaction instructions vary in specificity, and the method is designed to span this range. An underspecified instruction such as "a person lifting a table" fixes the interaction only at a high level and leaves the body configuration and contact layout to be inferred. A functional instruction such as "a person opening a washing machine" additionally requires reasoning that contact should occur at a functional region, near the door or handle, rather than anywhere on the object. A detailed instruction, such as the treadmill example above, constrains the pose and the individual body-part contacts explicitly. In every case the underlying problem is the same: the instruction must be converted into metric 3D contact constraints that are consistent with the reconstructed scene.

A natural approach to solve this problem is to inpaint a human performing the interaction into the input view and then recover that person's pose. This yields a body that looks correct in the image, but a pose recovered from a single view is expressed in image space and carries no guarantee of metric consistency with the scene: the recovered body may float above the floor, intersect furniture, or contact the wrong object region in 3D. The method's key idea is therefore to use the inpainted frame not only to recover the body but also to *localize* fine-grained scene-side contact regions, which are grounded on the reconstructed mesh and used to drive SMPL-X optimization so that the final body respects the scene geometry.

Figure 3.1 shows the complete pipeline, which comprises five modules with the data flow

$$\text{SIG} \;\rightarrow\; \text{human frame} \;\rightarrow\; \begin{Bmatrix} \text{contact estimation} \\ \text{pose initialization} \end{Bmatrix} \;\rightarrow\; \text{optimization.}$$

The first module converts the instruction and the view into a *Scene Interaction Graph* (SIG), which specifies the target object or floor, the relevant human body parts, and the required body-part–scene contact edges. The second module uses an image generation model to insert a single plausible human into the view according to the interaction instruction, producing a visual interaction hypothesis that captures pose, orientation, laterality, and approximate contact layout. This generated human frame then feeds two branches that run in parallel. In the first branch, the third module estimates scene-side contact regions: it marks each required contact on the original view with a distinct color to obtain body-part-specific contact masks, obtains floor-support regions separately with SAM3 when the interaction requires them, and projects the accepted masks onto the reconstructed mesh to form 3D contact targets. In the second branch, the fourth module initializes the body by passing the generated frame to GVHMR, which predicts an SMPL-X body conditioned on the camera focal length of the posed view. The fifth module refines this initialization through scene-aware optimization, holding the scene geometry and the projected contact targets fixed while optimizing the SMPL-X translation, orientation, pose, and a bounded global scale under a combination of contact attraction, penetration avoidance, and priors that keep the result close to the GVHMR estimate.

The full prompts used for the foundation-model stages are listed in appendix A.1, and the concrete model versions, hyperparameters, and iteration budgets are reported in the implementation details of **??**. This chapter describes the algorithmic role of each component; the appendix and the experiments chapter record the exact prompt wording and numerical

Methodology pipeline placeholder (treadmill running example)

posed view I + instruction y $\rightarrow$ SIG $\rightarrow$ human frame $\hat{I}$ $\rightarrow$ contact masks $\rightarrow$ 3D contact targets P_e^s $\rightarrow$ GVHMR init θ_0 $\rightarrow$ optimized body θ^*

Figure 3.1: Overview of the proposed zero-shot static HSI synthesis pipeline. Given a natural-language instruction, a posed ScanNet++ RGB view, and the reconstructed scene mesh, the method generates a Scene Interaction Graph, synthesizes a human interaction image, estimates body-part-specific scene contact regions on the original view, projects these regions onto the scene mesh, initializes an SMPL-X body with GVHMR, and refines it through scene-aware optimization.

settings.

3.2 Problem Formulation

Given a natural-language interaction instruction, a posed RGB view, and a reconstructed scene mesh, the goal is to synthesize a single human body that realizes the requested interaction in the scene coordinate frame. Let $I \in \mathbb{R}^{H \times W \times 3}$ denote the RGB view, which the user selects from the ScanNet++ scene. The view is calibrated: its intrinsics are given by the matrix K, and its pose in the scene coordinate frame is given by the rotation R_c and the translation t_c. The scene is represented by the triangle mesh $M_s = (V_s, F_s)$, with vertices $V_s \in \mathbb{R}^{N_s \times 3}$ expressed in the scene coordinate frame and triangular faces F_s.

The instruction y may range from underspecified to detailed, as in the examples of section 3.1. In all cases the method must infer a body configuration consistent with both the instruction and the scene geometry.

The output is a posed SMPL-X body expressed in the same coordinate frame as the reconstructed scene. SMPL-X is a differentiable parametric body model whose parameters are denoted

$$\theta = (\mathbf{T},\, \mathbf{r},\, \mathbf{p},\, \boldsymbol{\beta},\, s), \tag{3.1}$$

where $\mathbf{T}$ is the global translation, $\mathbf{r}$ the global orientation in axis-angle form, $\mathbf{p}$ the articulated body pose, $\boldsymbol{\beta}$ the body shape, and s a global isotropic scale applied to the body about its translation. Evaluating the model yields the human mesh $M_h(\theta) = (V_h(\theta), F_h)$

with posed vertices $V_h(\theta) \in \mathbb{R}^{N_h \times 3}$ and fixed faces F_h.

The synthesis target is a single static interaction state rather than a motion sequence. For object-manipulation instructions, the object remains in its reconstructed configuration; the method synthesizes the body that makes the interaction geometrically and visually plausible and does not simulate any subsequent object motion or state change.

The instruction is converted into a set of body–scene contact requirements, represented by a Scene Interaction Graph

$$G = (\mathcal{H}, \mathcal{S}, \mathcal{E}), \tag{3.2}$$

where $\mathcal{H}$ is the set of human body-part nodes, $\mathcal{S}$ the set of scene nodes, and $\mathcal{E}$ the set of contact edges. Human nodes are drawn from a fixed body-part vocabulary (hands, arms, feet, legs, hips, back, and head), and scene nodes are either the target object or the floor. Each edge

$$e = (h_e, s_e), \qquad e \in \mathcal{E}, \tag{3.3}$$

requires body part h_e to contact scene element s_e.

Each human node h_e maps to a fixed subset of SMPL-X contact vertices $B_e \subset \{1, \ldots, N_h\}$, defined once for the whole method. Under a body configuration θ, the corresponding body-side contact points are

$$P_e^h(\theta) = \{ V_h(\theta)_i \mid i \in B_e \}. \tag{3.4}$$

The scene-side contact target for edge e is not annotated manually. It is estimated from generated visual contact annotations and projected onto the reconstructed mesh, yielding a set of scene points $P_e^s \subset \mathbb{R}^3$.

The body is initialized from the generated human frame using GVHMR, giving an initial estimate $\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \boldsymbol{\beta}_0, s_0)$. During optimization, the global translation $\mathbf{T}$, global orientation $\mathbf{r}$, body pose $\mathbf{p}$, and global scale s are updated, while the shape $\boldsymbol{\beta}_0$ is taken from GVHMR and held fixed. Fixing $\boldsymbol{\beta}_0$ prevents the optimizer from distorting body identity to satisfy contact, while the single scale degree of freedom absorbs the residual overall size error inherent in monocular recovery; together, contact is explained through placement, pose, and a bounded change of overall size rather than through a change of body identity.

The synthesis problem is then to find SMPL-X parameters whose body-side contact points $P_e^h(\theta)$ approach the corresponding scene-side targets P_e^s, while the body remains physically plausible and avoids penetrating the scene. This is expressed as a scene-aware objective $\mathcal{L}_{\text{total}}$ that combines *data terms*, which tie the body to the observed scene through contact and penetration, with *regularization terms*, which keep the body plausible and close to the GVHMR initialization; both groups are defined in section 3.7. The final body is obtained by

$$\theta^* = \underset{\mathbf{T}, \mathbf{r}, \mathbf{p}, s}{\arg\min}\ \mathcal{L}_{\text{total}}(\theta), \qquad \boldsymbol{\beta} = \boldsymbol{\beta}_0. \tag{3.5}$$

The following sections describe how the graph G, the human initialization θ_0, and the scene-side contact targets P_e^s are obtained.

3.3 Scene Interaction Graph Generation

The first module converts the open-language instruction into an explicit, symbolic specification of the required contacts. Its purpose is not to solve the geometry of the interaction but to determine which human body parts should contact which scene elements. This intermediate representation, the Scene Interaction Graph (SIG), provides the structure that later guides both contact-region estimation and SMPL-X optimization. The prompt used for SIG generation is given in appendix A.1; the description here concerns only the functional role of the module and the structure of its output.

The SIG is generated from the instruction y and the posed view I. A vision-language model is prompted to identify the target object, the human body parts that are physically relevant to the interaction, and any required floor support, and to return two outputs: the graph $G = (\mathcal{H}, \mathcal{S}, \mathcal{E})$ of equation (3.2), and a detailed interaction description $\hat{y}$ that elaborates the original instruction. In the graph, each human node names a supported body part, each scene node names the target object or the floor, and each interaction edge pairs a human node with a scene node. The description $\hat{y}$ expands the terse instruction y into a fuller account of the interaction, specifying the pose and the individual per-human-body-part (per-hand and per-foot etc) contacts consistent with the graph edges, together with appearance attributes of the person such as clothing and hair. The graph fixes the symbolic contact structure used by contact estimation and optimization, while $\hat{y}$ conditions the image-generation stage that follows (section 3.4); the appearance attributes are included because they improve the realism and stability of the generated frame, on which the downstream pose recovery depends.

Body-part vocabulary. The human nodes are restricted to a fixed vocabulary: left and right hand, left and right arm, left and right leg, left and right foot, head, hips, and back. We fix this vocabulary so that every human node maps to a predefined SMPL-X contact region, a vertex subset $B_e \subset \{1, \ldots, N_h\}$ on the body mesh (figure 3.2). Hand nodes map to palm-side vertices, foot nodes to the sole-facing vertices, and sitting or leaning nodes to the corresponding regions on the hips, back, head, or legs. These subsets are defined once from the SMPL-X template and reused for all interactions. The restriction guarantees that every edge in a valid graph is grounded on one known body-side vertex set, which keeps the downstream optimization well defined and prevents the model from introducing body

SMPL-X contact-region placeholder

Fixed contact-vertex subsets B_e on the SMPL-X template,
color-coded per body-part node in the SIG vocabulary
(treadmill example activates left hand, right hand, left foot)

Figure 3.2: Predefined SMPL-X contact regions for the supported body-part vocabulary. Each body-part node in the Scene Interaction Graph maps to a fixed vertex subset B_e on the SMPL-X template, shown here color-coded. These subsets are defined once and reused for all interactions; the treadmill running example activates the left-hand, right-hand, and left-foot regions.

parts that cannot be attached to the body mesh.

Scene nodes and functional regions. Scene nodes are deliberately restricted to the target object and, when needed, the floor. The graph does not name object parts such as handle, knob, rung, or edge, and it does not attempt to localize them. An edge specifies only that a given body part contacts the target object; where on the object that contact should occur is left to the description $\hat{y}$ and to the later image-space estimation stage (section 3.5). This separation keeps the symbolic graph compact and robust: the language model is asked only for the coarse contact structure, which it can produce reliably, while the fine-grained localization of functional surfaces is deferred to a stage that has access to the rendered interaction and the calibrated view.

Table 3.2 shows the SIG produced for the treadmill running example. The instruction specifies both hands on the side handles and one foot on the belt, with the other leg lifted; the resulting graph contains two hand–treadmill edges and one foot–treadmill edge, and no floor edge, since the lifted leg provides no support and the supporting foot rests on the belt rather than the floor. The functional targets (handles, belt) appear nowhere in the graph itself; they are carried by the description $\hat{y}$ and localized later in image space. For the treadmill example, $\hat{y}$ reads, in part, "a person runs on the treadmill in a forward-moving pose, the left hand lightly holding the left side handle and the right hand the right side handle, the left foot landing on the belt while the right leg lifts behind," together with appearance attributes such as tied-back hair, an athletic top, leggings, and running shoes. This description drives the human-frame generation stage, while the graph in table 3.2 drives contact estimation and optimization. For contrast, a functional instruction such as “a person opening a washing machine” yields a single hand–washing-machine edge together with two foot–floor support edges, illustrating how the same structure spans instructions of differing specificity.

Validation. A generated graph is validated before it is used by later modules. The validation checks that the graph contains at least one scene node, that every human node belongs to the supported vocabulary, and that every edge references a defined scene node. Duplicate edges are merged, and edges with unsupported body parts are rejected. If no usable contact edge remains, the graph cannot yield meaningful contact targets and is regenerated. A validated graph therefore guarantees that each edge maps to exactly one body-side contact set B_e and one scene node, ready for the contact-estimation and optimization stages.

In summary, the SIG links language to geometry in a controlled way. It fixes which body parts participate and which scene element each of them contacts, and it produces an elaborated description $\hat{y}$ of the interaction, but it deliberately leaves two things unresolved: the exact object surface to be touched, and the final body pose. The next module uses $\hat{y}$ to generate a visual hypothesis of the interaction, from which the following modules localize the scene-side contact regions and initialize the body.

3.4 Human Frame Generation

The second module produces a visual hypothesis of the requested interaction in the input view. Given the posed view I and the detailed interaction description $\hat{y}$ from the SIG, an image generation model inpaints a single human into the scene so that the person appears to perform the interaction while the scene itself is left unchanged. The result is the *human frame*, an image in the same coordinate system as I that shows the hypothesized pose, orientation, laterality, and approximate contact layout. This frame is the shared input to the two parallel branches of the pipeline: it provides the image evidence from which GVHMR initializes the body (section 3.6), and it shows the interaction from which the scene-side contact regions are estimated (section 3.5). The prompts used in this module are given in appendix A.1.

Because both branches derive from the same human frame, the initialized body and the estimated contacts describe one and the same hypothesized interaction. This shared origin is what makes the two consistent when they are later combined: the contact targets and the initial pose are not independent estimates that must be reconciled, but two readings of the same generated evidence.

Preserving the scene. The generation prompt is constructed to keep the input view intact apart from the inserted person. The camera viewpoint, room geometry, object layout, lighting, textures, and target object are all to be preserved, and exactly one person is inserted. For object-manipulation instructions, the target object must not be moved, opened, lifted, rotated, or deformed. This constraint matters because the reconstructed mesh M_s represents the object in its original configuration and cannot reflect an image-only edit: for "a person opening a washing machine," the desired frame shows the person reaching toward the closed door or handle, not a washing machine whose door has been drawn open. For the treadmill example, the constraint keeps the treadmill in place and undeformed while the running person is inserted upon it.

Respecting the contact structure. The human frame must also be consistent with the contacts encoded in the SIG. Hand–object edges call for the corresponding hands to be placed near the target object, foot–floor edges call for the person to appear supported by the floor, and asymmetric contacts must be preserved with the correct laterality. For the treadmill example, the frame should show both hands on the side handles, the left foot on the belt, and the right leg lifted behind, matching the graph and the description $\hat{y}$. Preserving this structure is what makes the frame a useful bridge between the symbolic interaction and the metric optimization that follows.

Candidate selection. The image model produces a set of candidate frames

$$\mathcal{I}_H = \{ \hat{I}^{(1)}, \hat{I}^{(2)}, \dots, \hat{I}^{(N)} \}, \tag{3.6}$$

from which a vision-language evaluator selects the most suitable one. The evaluator compares each candidate against the input view I, and the detailed instruction $\hat{y}$, and favors candidates that express the interaction clearly, preserve the target object and surrounding scene, exhibit a plausible full-body pose, and expose the relevant body-part contacts for later reasoning. Candidates with severe object distortion, missing body parts, implausible body scale, incorrect laterality, or unrelated interaction geometry are rejected. The selected frame is

$$\hat{I} = \hat{I}^{(j^\star)}. \tag{3.7}$$

Minor visual artifacts in $\hat{I}$ are tolerated, since the frame serves as evidence for pose and contact reasoning rather than as a final rendering; the selection step exists to ensure that both downstream branches operate on a frame that is visually consistent with the requested interaction. The number of candidates and the selection prompt are reported in **??** and appendix A.1, respectively.

The selected human frame $\hat{I}$ provides the visual evidence used by the following modules, while the authoritative scene geometry remains the reconstructed mesh M_s and the

Human frame generation placeholder (treadmill running example)

posed view I $\rightarrow$ candidate frames $\mathcal{I}_H$ $\rightarrow$ selected frame $\hat{I}$

Figure 3.3: Human frame generation and selection. The image model inpaints one person into the posed ScanNet++ view according to the interaction description $\hat{y}$, preserving the scene, camera viewpoint, and target object. A vision-language evaluator then selects the candidate that best preserves the scene and expresses the required body–scene contacts. The selected frame $\hat{I}$ is the shared input to GVHMR initialization and contact estimation.

authoritative scene appearance for contact annotation remains the original view I. The next section describes how the contact regions implied by $\hat{I}$ are localized on I and grounded on M_s.

3.5 Agentic Contact Estimation

The third module estimates the scene-side contact regions required by the interaction and grounds them on the reconstructed mesh. Given the human frame $\hat{I}$, the posed view I, and the graph G, it produces, for each contact edge, a verified 2D contact mask on the view I and projects it to the mesh to obtain the scene-side target P_e^s. This module is the bridge between the open-vocabulary visual reasoning of the foundation models and the metric optimization that follows: it is where the functional contact regions left unresolved by the SIG (the handles and the belt, in the treadmill example) are finally localized in the calibrated image and lifted into 3D.

The graph G re-enters the pipeline at this point. While human-frame generation is driven by the elaborated description $\hat{y}$, contact estimation is driven by the graph, since it needs the explicit per-edge structure: one mask, and later one 3D target, per contact edge.

Two distinct difficulties must be overcome, and the module addresses them by separate means. First, image generation is a powerful way to *reason* about where contact occurs but an unreliable way to *record* it: asked to paint contact regions onto the scene, an image model may silently alter the object, the background, or the image boundaries, and such drift would be fatal here, since the masks are projected to the mesh through the calibrated camera and must correspond pixel-for-pixel to the view I. This is handled structurally, by using the image model only to propose colored regions, extracting those colored pixels, and compositing them onto the untouched view I; the resulting *contact composite* carries the proposed contacts while retaining, by construction, the exact appearance and geometry of the calibrated view. Second, even on an untouched canvas the localization itself may be wrong: a required contact may be missing, a spurious one may appear, a mask may sit on the wrong object surface, or a footprint may be misplaced or wrongly sized. This is handled procedurally, by a closed-loop vision-language evaluator that inspects each composite and, when necessary, issues a correction for another round. The first mechanism guarantees geometric fidelity of the annotated image; the second, the *agentic* verify-and-correct loop, guarantees the correctness of the contacts drawn on it.

3.5.1 Contact-Overlay Generation

Contact estimation uses two images in complementary and authoritative roles. The human frame $\hat{I}$ is the *reference*: it is used solely to infer which body parts contact the target object and where those contacts fall on the object surface. The posed view I is the *canvas*: it is the base image and the authority for the target object's position, geometry, and boundaries. This division is enforced throughout the module: the reference determines *where contact occurs*, and the canvas determines *where the object is*, so that masks are always aligned to the geometry the projection step will use.

Each object-contact edge $e \in \mathcal{E}_{\text{obj}}$ is assigned a distinct, saturated color c_e, where $\mathcal{E}_{\text{obj}}$ is the subset of edges whose scene node is the target object. For the treadmill example, the left-hand, right-hand, and left-foot edges receive three different colors, so that their masks can be separated unambiguously after generation. At round r, the contact-overlay model receives the reference, the canvas, the graph, the color assignment, the previous composite $C^{(r-1)}$, and the correction instruction $a^{(r-1)}$, and produces an overlay candidate

$$O^{(r)} = \Phi_{\text{con}}\big(\hat{I},\, I,\, G,\, \{c_e\},\, C^{(r-1)},\, a^{(r-1)}\big), \tag{3.8}$$

where Φ_{con} denotes the contact-overlay generation model. The first round uses neither a previous composite nor a correction instruction. When they are present, the model preserves the masks the evaluator has already accepted, copying them forward from $C^{(r-1)}$, and redraws only the regions the correction instruction identifies; this prevents already-correct masks from being regenerated and drifting between rounds.

Each mask is drawn as the localized *contact footprint* on the object surface, representing the apparent extent of contact rather than the silhouette of the touching body part, and every mask pixel is clipped strictly to the target-object boundary in the canvas, so that no mask extends onto the person, floor, wall, or background. The intended $O^{(r)}$ is thus a copy of the canvas bearing only these solid-color footprints; because it may nonetheless carry unintended edits to the scene, it is treated only as a source of colored regions, from which the masks are extracted next. The overlay prompt is given in appendix A.1.

3.5.2 Mask Extraction and Composite Construction

For each object-contact edge, the pixels matching the assigned color are extracted from the overlay candidate, yielding a binary mask

$$Z_e^{(r)} = \mathrm{ExtractColor}\big(O^{(r)},\, c_e\big). \tag{3.9}$$

Matching is performed within a small color tolerance, and small isolated components may be discarded. This step keeps only the painted footprints and discards everything else in $O^{(r)}$, including any incidental changes to object shape, texture, lighting, or background.

The extracted masks are then composited onto the untouched view I, producing the contact composite

$$C^{(r)} = \mathrm{Composite}\big(I,\, \{Z_e^{(r)}\}\big). \tag{3.10}$$

By construction, $C^{(r)}$ shares the resolution, camera calibration, and scene appearance of I, and differs from it only by the added contact overlays. This is what makes the composite, rather than the raw overlay, the authoritative annotated image: the image model is used only for its strength of proposing where contact occurs, while the calibrated view remains the sole geometric authority, and any scene drift in $O^{(r)}$ is discarded with the unmatched pixels.

3.5.3 Agentic Verification and Correction

The contact composite is verified by a vision-language evaluator in a closed loop. The evaluator receives the reference $\hat{I}$, the canvas I, and the current composite $C^{(r)}$: the reference is authoritative for which body parts contact the object and where, the canvas is authoritative for the object geometry, and the composite carries the masks under review. Crucially, a mask is not accepted merely because it lies plausibly on the object; it is accepted only if its location matches the contact shown in the reference, mapped onto the same object region in the canvas. Against this standard the evaluator checks for the characteristic failure modes: a missing required contact, an extra contact, a wrong body-part color, a mask on the wrong object surface, a mask shifted from the reference location, and a footprint of the wrong size. It returns a decision together with a natural-language correction instruction,

$$\big(d^{(r)},\, a^{(r)}\big) = \Phi_{\mathrm{eval}}\big(\hat{I},\, I,\, C^{(r)},\, G,\, \{c_e\}\big), \tag{3.11}$$

where $d^{(r)}$ is the accept/reject decision and $a^{(r)}$ is a single consolidated instruction covering every mask to be corrected; for example, moving a hand mask onto the correct handle, shrinking an oversized footprint, adding a missing contact, or removing a spurious one. Acceptance requires that all required contacts are present, no extra masks exist, every color is correct, and every footprint matches the surface, location, and size indicated by the reference. The evaluation prompt is given in appendix A.1.

On acceptance, the loop terminates and the current composite is used. On rejection, the correction instruction and the current composite are passed to the overlay model for the next round, giving the cycle

$$O^{(r)} \;\to\; \{Z_e^{(r)}\} \;\to\; C^{(r)} \;\to\; \big(d^{(r)}, a^{(r)}\big) \;\to\; O^{(r+1)}. \tag{3.12}$$

The loop runs for a fixed maximum number of rounds. It stops as soon as the evaluator accepts a composite; if the maximum is reached without acceptance, the masks from the final round are taken as the result, so that the procedure always returns a usable annotation. The maximum number of rounds is reported in **??**.

Let $r^\star$ denote the selected round. The final object-contact masks are

$$\Omega_e^{2D} = Z_e^{(r^\star)}, \qquad e \in \mathcal{E}_{\text{obj}}. \tag{3.13}$$

3.5.4 Floor-Support Masks

Floor contacts are handled separately from object contacts. For a support edge such as a foot on the floor, the precise pixel footprint matters less than a geometrically valid support surface, and asking the overlay model to paint a footprint on an empty stretch of floor invites hallucinated placement. Instead, SAM3 segments the visible floor in the view I, and the resulting region serves as the scene-side support surface; the optimizer then determines the exact foot placement on it, subject to the contact, penetration, and pose terms of section 3.7.

Let $\mathcal{E}_{\text{floor}}$ denote the edges whose scene node is the floor, and let $\Omega_{\text{floor}}^{2D}$ be the segmented floor mask. Each floor-support edge takes this mask as its 2D contact region,

$$\Omega_e^{2D} = \Omega_{\text{floor}}^{2D}, \qquad e \in \mathcal{E}_{\text{floor}}, \tag{3.14}$$

and the same mask may be shared across several foot-support edges, since the distinction between left and right foot is enforced on the body side during optimization. The treadmill example has no floor edge, as its supporting foot rests on the belt; an interaction such as a person standing to open a washing machine, whose graph includes two foot–floor edges, would use this floor mask for both.

3.5.5 Projection to 3D Contact Targets

The verified 2D masks are lifted to the reconstructed mesh through the calibrated camera. A scene point $x \in \mathbb{R}^3$ in the scene coordinate frame projects to the image as

$$\tilde{u} = K\big(R_c x + t_c\big), \tag{3.15}$$

followed by perspective division to give the pixel coordinate u; write this calibrated projection as $\pi(x; K, R_c, t_c)$. A scene vertex or sampled surface point is assigned to edge e when its projection falls inside the mask Ω_e^{2D} and it is visible in the view. The scene-side contact target for edge e is therefore

$$P_e^s = \big\{\, x \in M_s \;\big|\; \pi(x; K, R_c, t_c) \in \Omega_e^{2D} \text{ and } x \text{ visible} \,\big\}, \tag{3.16}$$

represented in practice as a set of scene vertices or sampled surface points. These points are fixed for the remainder of the pipeline and act as the regions that the corresponding SMPL-X contact vertices are drawn toward. Collecting the targets over all edges gives the output of the module,

$$\mathcal{P}^s = \big\{\, P_e^s \;\big|\; e \in \mathcal{E} \,\big\}, \tag{3.17}$$

in which each element is linked, through its edge, to one body-side contact set B_e. The next module initializes the SMPL-X body, after which optimization brings the two sides into contact.

3.6 SMPL-X Pose Initialization

The fourth module initializes the SMPL-X body from the human frame. This is the second of the two branches fed by $\hat{I}$: while contact estimation localizes the scene-side targets, this module recovers a complete articulated body to place against them. The scene-side targets constrain only where a few body regions should end up and do not by themselves determine a full pose. A good initial body is therefore essential, and not merely convenient: the scene-aware objective of section 3.7 is non-convex, so the quality of its solution depends heavily on where it starts. An initialization close to the intended interaction lets the optimizer make small, local adjustments, whereas a naive or arbitrary starting body would leave it prone to poor local minima. The method obtains this initialization with GVHMR.

GVHMR performs monocular human recovery, predicting an SMPL-X body in the camera frame. As it is trained for this task, the body it recovers is a realistic, well-articulated pose that already captures the coarse configuration of the interaction shown in the human frame: the bend of the limbs, the global orientation, and the overall body arrangement are close to correct before any scene-aware refinement. This is precisely the property the optimization needs. Because the initial body already sits near a plausible solution, the

Agentic contact estimation placeholder (treadmill running example)

reference $\hat{I}$ + canvas I $\rightarrow$ overlay $O^{(r)}$ $\rightarrow$ extracted masks $Z_e^{(r)}$ $\rightarrow$ composite $C^{(r)}$

evaluator: accept or correct $\rightarrow$ final masks Ω_e^{2D} $\rightarrow$ 3D targets P_e^s

Figure 3.4: Agentic contact estimation. Using the human frame $\hat{I}$ as reference and the posed view I as canvas, the overlay model paints colored contact footprints, of which only the colored pixels are extracted and composited back onto the untouched view I. A vision-language evaluator checks the composite against the reference and the canvas and either accepts it or returns a correction for another round. Accepted masks are projected through the calibrated camera to the reconstructed mesh to yield body-part-specific 3D contact targets P_e^s. For the treadmill example, the left hand, right hand, and left foot receive distinct colors and are localized on the two handles and the belt.

subsequent refinement adjusts placement and pose only locally, which keeps it clear of the poor local minima that a weaker initialization would invite.

Two calibration details keep this initialization metrically consistent with the rest of the pipeline. First, GVHMR is conditioned on the camera focal length, which is read from the intrinsics K of the posed view; supplying the true focal length rather than a default keeps the recovered body consistent with the calibrated camera. Second, the camera-frame estimate is transformed into the scene coordinate frame using the extrinsics (R_c, t_c) of the same view, so that the initial body is placed in the same frame as the reconstructed mesh and the projected contact targets, expressed through the very geometry used to project those targets.

Since the setting provides a single static frame rather than a video, the frame is repeated to form a short static clip,

$$\mathcal{V}_{\hat{I}} = \big(\hat{I},\, \hat{I},\, \dots,\, \hat{I}\big), \tag{3.18}$$

which GVHMR processes as an ordinary sequence; the first recovered frame is taken as the initialization. Repeating the frame preserves the static nature of the target interaction while satisfying the model's sequence input.

Applying GVHMR to the clip yields an initial SMPL-X estimate in the camera frame,

$$\theta_0^{\text{cam}} = \big(\mathbf{T}_0^{\text{cam}},\, \mathbf{r}_0^{\text{cam}},\, \mathbf{p}_0,\, \boldsymbol{\beta}_0\big), \tag{3.19}$$

with global translation $\mathbf{T}_0^{\text{cam}}$ and orientation $\mathbf{r}_0^{\text{cam}}$ expressed relative to the camera, articulated pose $\mathbf{p}_0$, and shape $\boldsymbol{\beta}_0$. The pose and shape are frame-relative rather than camera-relative and are carried over unchanged; only the global translation and orientation must be moved into the scene coordinate frame.

The optimization runs in the scene frame, so the initialization is transformed from camera to world coordinates using the pose of the selected view. With the convention

$$x^{\text{cam}} = R_c \, x^{\text{world}} + t_c, \tag{3.20}$$

a camera-frame point maps to the world frame by the inverse transform. The global translation is converted as

$$\mathbf{T}_0 = R_c^{\top} \big(\mathbf{T}_0^{\text{cam}} - t_c \big), \tag{3.21}$$

and the global orientation by applying the inverse camera rotation. Writing $\mathbf{R}(\cdot)$ for the rotation matrix of an axis-angle vector, the world-frame orientation matrix is

$$\mathbf{R}(\mathbf{r}_0) = R_c^{\top} \, \mathbf{R}\big(\mathbf{r}_0^{\text{cam}}\big), \tag{3.22}$$

whose axis-angle form is $\mathbf{r}_0$. The initialization in the scene frame is then

$$\theta_0 = \big(\mathbf{T}_0, \, \mathbf{r}_0, \, \mathbf{p}_0, \, \boldsymbol{\beta}_0, \, s_0\big), \tag{3.23}$$

where the global scale is initialized to $s_0 = 1$.

While the recovered pose is realistic, it is estimated from the human frame alone and is therefore unaware of the reconstructed scene: it is not guaranteed to satisfy the projected contact targets or to avoid intersecting the mesh, and the frame may carry small errors in scale, depth, or contact placement. The role of the final module is to refine the translation $\mathbf{T}_0$, orientation $\mathbf{r}_0$, pose $\mathbf{p}_0$, and global scale s_0 against the scene geometry and the contact targets, starting from this strong initialization. Throughout that refinement the shape $\boldsymbol{\beta}_0$ is held fixed, so that identity is preserved: contact is explained by placement, pose, and a bounded overall scale rather than by altering body identity, while the single scale degree of freedom absorbs the residual size error of the monocular estimate. The reconstructed mesh and the projected targets likewise remain fixed.

3.7 Scene-Aware SMPL-X Optimization

The fifth module refines the GVHMR initialization into the final body. The initialization is a plausible, well-articulated pose, but it is unaware of the scene: it need not satisfy the projected contact targets, and it may intersect the reconstructed mesh. This module corrects those errors by minimizing a scene-aware objective while holding the scene fixed.

The objective combines two *data terms*, which tie the body to the observed scene through contact and non-penetration, with four *regularization terms*, which keep the body plausible and anchored to the trusted initialization. The scene mesh M_s and the projected contact targets $\{P_e^s\}$ remain fixed throughout; only the body is optimized.

3.7.1 Optimized Variables

The optimization starts from the world-frame initialization $\theta_0 = (\mathbf{T}_0, \mathbf{r}_0, \mathbf{p}_0, \boldsymbol{\beta}_0, s_0)$ of section 3.6 and updates the global translation $\mathbf{T}$, global orientation $\mathbf{r}$, articulated pose $\mathbf{p}$, and global scale s, while the shape $\boldsymbol{\beta}_0$ is held fixed:

$$\theta = (\mathbf{T},\, \mathbf{r},\, \mathbf{p},\, \boldsymbol{\beta}_0,\, s). \tag{3.24}$$

Fixing $\boldsymbol{\beta}_0$ preserves body identity, so that contact is achieved through placement, articulation, and a bounded change of overall size rather than by distorting the body's proportions. The global orientation is optimized through a continuous six-dimensional rotation representation, which avoids the discontinuities of the axis-angle parameterization and yields a stable gradient for the orientation update. For any parameter state, the SMPL-X model produces the posed mesh $M_h(\theta) = (V_h(\theta), F_h)$, whose contact vertices are compared against the fixed scene targets.

3.7.2 Data Terms

Contact objective. The contact term attracts each body-side contact region toward its projected scene-side target. For edge e, the body-side contact points $P_e^h(\theta) = \{V_h(\theta)_i \mid i \in B_e\}$ are pulled toward the target P_e^s by a mean one-sided nearest-neighbor distance:

$$\mathcal{L}_{\text{contact}}(\theta) = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{1}{|B_e|} \sum_{i \in B_e} \min_{q \in P_e^s} \left\| V_h(\theta)_i - q \right\|_2^2. \tag{3.25}$$

The distance is measured in one direction only, from the body to the scene: each body contact vertex is drawn to its nearest point in the scene target, but scene points are under no obligation to be reached. This direction reflects the geometry of the problem. The body-side set B_e is a small, fixed region whose every vertex should make contact, whereas the scene-side target P_e^s is a broad projected region of which only a part is actually touched, since a palm covers a small patch of a handle and a sole a small patch of the belt. A symmetric distance, which would also pull each scene point toward the body, would therefore penalize the uncovered remainder of the target and try to spread the body across the entire region rather than let it rest naturally on a portion of it. Matching only from the body to the scene also removes any need for a fixed body-to-scene vertex correspondence and lets the target act as a region the contact vertices may settle anywhere within. Averaging over the whole set B_e, rather than taking a single closest vertex, then encourages the entire contact region to meet the surface: for the treadmill example, this draws the full palms toward the two handles and the sole toward the belt.

Penetration objective. The contact term alone can pull the body through the scene surface, so a penetration term discourages the body from occupying scene geometry. Using VolumetricSMPL [28], which equips the posed SMPL-X body with a differentiable signed-distance field, the method queries the signed distance $\phi_\theta(x)$ from each of a set of scene points $\mathcal{Q}_s$, sampled from the reconstructed mesh near the body, to the body surface, with $\phi_\theta(x) < 0$ indicating that x lies inside the body. The term penalizes scene points that fall inside the body or within a small clearance margin $\delta \geq 0$:

$$\mathcal{L}_{\text{pen}}(\theta) = \frac{1}{|\mathcal{Q}_s|} \sum_{x \in \mathcal{Q}_s} \operatorname{ReLU}\big(\delta - \phi_\theta(x)\big). \tag{3.26}$$

Points comfortably outside the body contribute nothing, so the term acts only where the body would otherwise intersect the scene, allowing contact while forbidding interpenetration.

3.7.3 Regularization Terms

Pose prior. Because the GVHMR pose is already realistic, the articulated pose is kept close to its initial value, so that contact is satisfied by small adjustments rather than by drifting into an unnatural configuration:

$$\mathcal{L}_{\text{pose}}(\theta) = \|\mathbf{p} - \mathbf{p}_0\|_2^2. \tag{3.27}$$

Orientation prior. The global orientation is likewise anchored to the initialization, penalized by the geodesic distance between the current and initial rotations:

$$\mathcal{L}_{\text{orient}}(\theta) = \big\|\operatorname{Log}\!\big(\mathbf{R}(\mathbf{r}_0)^\top\, \mathbf{R}(\mathbf{r})\big)\big\|_2^2, \tag{3.28}$$

where $\mathbf{R}(\cdot)$ is the rotation matrix of an axis-angle vector and $\operatorname{Log} : SO(3) \to \mathbb{R}^3$ returns the axis-angle vector of a rotation. This prevents the optimizer from satisfying contact through large, unrealistic global rotations inconsistent with the human frame.

Height prior. The global scale s is free to correct the overall size of the monocular estimate, but it must not drive the body to an implausible stature. Writing the body's metric height as $h(\theta) = s\, h_{\text{can}}$, where h_{can} is the height of the fixed-shape body at unit scale, the height prior keeps this stature near a target h_{tgt}:

$$\mathcal{L}_{\text{height}}(\theta) = \left(\frac{h(\theta) - h_{\text{tgt}}}{\sigma_h} \right)^2, \tag{3.29}$$

with σ_h setting the tolerated deviation. Together with a bounded parameterization of the scale (**??**), this term confines s to a realistic human height range, so that the size correction resolves the residual scale error of monocular recovery without introducing an implausibly small or large body.

Self-intersection prior. Finally, a self-intersection term discourages the body mesh from penetrating itself, for instance an arm passing through the torso. Following [29], the colliding triangles of the posed body are first detected with a bounding-volume hierarchy [30], giving a set $\mathcal{C}(\theta)$ of intersecting triangle pairs; a local conic distance-field penalty ψ then measures the interpenetration of each colliding pair and pushes them apart:

$$\mathcal{L}_{\text{self}}(\theta) = \frac{1}{|\mathcal{C}(\theta)|} \sum_{(a,b)\in\mathcal{C}(\theta)} \psi\big(T_a(\theta),\, T_b(\theta)\big), \tag{3.30}$$

where $T_a(\theta)$ and $T_b(\theta)$ are the posed triangles of a colliding pair and ψ vanishes when they are disjoint. Only colliding pairs contribute, so the term is active solely where the body would otherwise intersect itself, keeping the refined body physically coherent without affecting collision-free regions. The parameters of ψ are given in **??**.

3.7.4 Total Objective

The full objective is the weighted sum of the two data terms and the four regularization terms:

$$\mathcal{L}_{\text{total}}(\theta) = \underbrace{\lambda_{\text{contact}}\mathcal{L}_{\text{contact}} + \lambda_{\text{pen}}\mathcal{L}_{\text{pen}}}_{\text{data terms}} + \underbrace{\lambda_{\text{pose}}\mathcal{L}_{\text{pose}} + \lambda_{\text{orient}}\mathcal{L}_{\text{orient}} + \lambda_{\text{height}}\mathcal{L}_{\text{height}} + \lambda_{\text{self}}\mathcal{L}_{\text{self}}}_{\text{regularization terms}}, \tag{3.31}$$

and the final body is obtained by

$$\theta^* = \underset{\mathbf{T},\mathbf{r},\mathbf{p},s}{\arg\min}\, \mathcal{L}_{\text{total}}(\theta), \qquad \boldsymbol{\beta} = \boldsymbol{\beta}_0. \tag{3.32}$$

The data terms enforce the functional body–scene relationships inferred by the earlier modules while forbidding interpenetration, and the regularization terms keep the result close to the trusted initialization and within a plausible human range. The loss weights $\lambda_{(\cdot)}$ are reported in **??**.

3.7.5 Optimization Procedure

The objective is minimized with Adam in two stages, following a coarse-to-fine schedule that exploits the quality of the initialization. In the first, *rigid* stage, only the global translation and scale are optimized, while the orientation and articulated pose are frozen; this places and sizes the trusted GVHMR body against the scene without deforming its pose, and the self-intersection term is disabled since the pose cannot change. In the second, *pose* stage, all variables, translation, orientation, pose, and scale, are optimized under the full objective, allowing local articulation to refine the contacts. Separating placement from articulation keeps the optimizer near the plausible initialization for as long as possible and reduces the risk of settling in a poor local minimum.

At each iteration the SMPL-X mesh is recomputed, the body contact vertices are matched to their fixed scene targets, penetration is evaluated against the sampled scene points, and gradients are backpropagated through the differentiable body model; the scene mesh, the projected contact targets, and the body shape stay fixed. The optimizer returns the final parameters

$$\theta^* = (\mathbf{T}^*, \mathbf{r}^*, \mathbf{p}^*, \boldsymbol{\beta}_0, s^*), \tag{3.33}$$

and the final synthesized interaction is the posed mesh

$$M_h^* = M_h(\theta^*), \tag{3.34}$$

already expressed in the ScanNet++ scene coordinate frame. The stage lengths, learning rate, gradient clipping, and total iteration budget are reported in **??**.

Table 3.1: Notation used throughout the methodology chapter.

Symbol	Meaning
Inputs and scene	
I	Posed RGB view selected by the user from a ScanNet++ scene.
K	Camera intrinsic matrix.
R_c, t_c	Camera rotation and translation (extrinsics) in the scene frame.
$M_s = (V_s, F_s)$	Reconstructed scene mesh in the scene coordinate frame.
y	Natural-language interaction instruction.
$\hat{y}$	Detailed interaction description generated with the SIG.
Scene Interaction Graph	
$G = (\mathcal{H}, \mathcal{S}, \mathcal{E})$	Scene Interaction Graph.
$e = (h_e, s_e)$	Contact edge between body part h_e and scene element s_e.
$\mathcal{E}_{\text{obj}}$, $\mathcal{E}_{\text{floor}}$	Object-contact and floor-support edge subsets.
Body and contact	
$\theta = (\mathbf{T}, \mathbf{r}, \mathbf{p}, \boldsymbol{\beta}, s)$	SMPL-X parameters: translation, orientation, pose, shape, scale.
$M_h(\theta) = (V_h(\theta), F_h)$	Posed SMPL-X mesh.
B_e	Fixed SMPL-X contact-vertex subset for the body part in edge e.
$P_e^h(\theta)$	Body-side contact points for edge e.
P_e^s	Scene-side contact target for edge e, on the reconstructed mesh.
Ω_e^{2D}	Verified 2D contact mask for edge e in the view I.
Foundation-model operators and intermediates	
$\hat{I}$	Selected generated human frame.
$\mathcal{I}_H$	Set of candidate human frames.
Φ_{con}, Φ_{eval}	Contact-overlay generation and contact-evaluation models.
$O^{(r)}$, $Z_e^{(r)}$, $C^{(r)}$	Overlay candidate, extracted mask, and composite at round r.
$\phi_\theta(x)$	Signed distance from scene point x to the posed body.
$\mathcal{Q}_s$	Scene points sampled near the body for penetration evaluation.
Optimization	
θ_0	GVHMR initialization in the scene frame.
$\mathcal{L}_{\text{total}}$	Scene-aware objective (section 3.7).
θ^*	Final optimized body parameters.

Table 3.2: Scene Interaction Graph for the treadmill running example. The graph records only the coarse body-part–object contacts; the functional target region (handle, belt) is carried by the description $\hat{y}$ and localized later in image space, rather than encoded as a graph node.

Human node	Scene node
left hand	treadmill
right hand	treadmill
left foot	treadmill

Chapter 4

Experiments

This chapter defines the experimental protocol used to validate the thesis claims.

4.1 Research Questions

State the questions the experiments answer. Each question should map to a table, figure, ablation, or qualitative analysis.

4.2 Datasets

Describe all datasets, splits, annotations, preprocessing steps, and any generated data.

4.3 Baselines

List baseline methods and explain how they were configured. Make comparisons fair and reproducible.

4.4 Metrics

Define quantitative metrics and explain what each metric captures. Include limitations of each metric where relevant.

Table 4.1: Example experiment configuration table. Replace with the final setup.

Item	Value
Dataset split	train / validation / test
Batch size	–
Optimizer	–
Hardware	–

4.5 Experimental Setup

Specify training schedule, optimizer, learning rate, batch size, number of runs, random seeds, hardware, and software versions.

Chapter 5

Results and Discussion

This chapter reports the main results, ablations, qualitative examples, and limitations.

5.1 Main Results

Present the primary quantitative results first. Explain the trend, not just the numbers.

5.2 Ablation Studies

Isolate the effect of each contribution. Keep ablations tied to the claims made in the introduction.

5.3 Qualitative Results

Show representative successes and failures. Qualitative figures should make the method's behavior inspectable.

5.4 Limitations

Discuss failure cases, data bias, compute cost, assumptions, and evaluation gaps. Strong limitations make the thesis more credible.

Table 5.1: Main result table placeholder.

Method	Metric 1	Metric 2	Metric 3
Baseline	–	–	–
Proposed	–	–	–

5.5 Discussion

Connect the results back to the motivation. Explain what the experiments imply for the broader research problem.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

Summarize the thesis in a few paragraphs: problem, approach, evidence, and main takeaway. Avoid introducing new experiments here.

6.2 Future Work

Describe concrete extensions that follow from the limitations and results. Separate near-term engineering improvements from longer-term research directions.

6.3 Closing Remarks

End with a brief statement about the expected impact or usefulness of the work.

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Appendix A

Additional Material

A.1 Foundation Model Prompts

This appendix records the system prompts used by the foundation-model stages in chapter 3. The symbols below are used in the methodology chapter.

Table A.1: System-prompt symbols used in the methodology.

Symbol	Stage	Source or status
Π_{SIG}	SIG generation	`4DHSI/01_Generate_SIG/system_prompt_sig.md`
Π_{H}	Human-frame inpainting	`4DHSI/02_Generate_Human_Frame/system_prompt_human_inpaint.md`
Π_{sel}	Best human-frame selection	Placeholder; final system prompt not fixed yet.
Π_{con}	Contact overlay generation	`4DHSI/03_Estimate_Contact_Agentic/system_prompt_estimate_contact_agentic.md`
Π_{eval}	Contact VLM evaluation	`4DHSI/03_Estimate_Contact_Agentic/vlm_prompt_evaluate_contact.md`

A.1.1 SIG Generation System Prompt Π_{SIG}

```
You generate a Scene Interaction Graph (SIG) for a static human-scene interaction.

Produce a compact graph containing the target object or objects, relevant human
body parts, direct physical contact edges, and a static interaction description.

Input:
- A scene image from the selected camera view.
- A JSON request:
```

```
{
  "interaction": "short interaction description"
}

Output only valid JSON. Do not include markdown or commentary.

Required output schema:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "short primary target object name"
    },
    {
      "id": "target_object_2",
      "label": "short secondary target object name"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "brief reason for this contact"
    }
  ],
  "interaction": "Static interaction description under 150 words."
}

Rules:

1. Use the interaction text to identify the intended human-scene contact, and use the
↪  image only to understand visible object context.
2. `target_objects` should contain the scene object that receive direct human contact
↪  in the static pose.
3. Use one target object when the contact is with a single object or multiple
↪  structural parts of the same object, such as a chair back, bed frame, or bathtub
↪  side.
4. Use two target objects only when two separate scene objects are directly contacted.
↪  For example, use "bathtub" and "curtain rail" when one foot contacts the bathtub
↪  and one hand contacts the curtain rail.
5. Keep target object labels short noun phrases.
6. Use human part nodes only from this vocabulary:
   - left hand
   - right hand
   - left arm
   - right arm
   - left leg
   - right leg
   - left foot
   - right foot
   - head
   - hips
   - back
```

```
7. Include a human part node only when that part is physically important for contact or pose.
8. `scene_nodes` can include only:
   - target_object_1
   - target_object_2
   - floor
9. Include `target_object_2` only if it exists in `target_objects` and has at least one direct contact edge.
10. `interaction_edges` represent direct physical contact only.
11. Use `scene_element: "target_object_1"` or `scene_element: "target_object_2"` for contacts with the corresponding target object.
12. Use floor edges only for required direct floor support. If there is no floor contact, simply omit `floor` from `scene_nodes` and `interaction_edges`.
13. Do not create object-part nodes.
14. Keep interaction edges focused on physically necessary contacts.
15. The output `interaction` must describe the static pose and physical contacts under 150 words. It must explicitly restate every `interaction_edges` contact using the same side-specific human part names, such as "left hand", "right foot", or "back", and the contacted scene object label or floor. Avoid vague phrases such as "one hand", "both feet", or "legs" when the edge uses specific left/right parts. It should also describe clothing, shoes, hair, and facial expression.

Few-shot examples:

These few-shot examples are written without a scene image. They demonstrate the
expected JSON structure and explicit contact wording. For the real request, use
the provided scene image to decide the exact contact configuration, including
which side-specific hand, foot, leg, or arm is in contact.

Example 1 input JSON:

{
  "interaction": "a person lifting the brown small wooden table with both hands"
}

Example 1 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "brown small wooden table"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the table while preparing to lift"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",
```

```
      "notes": "the right hand grips the opposite side of the table"
    },
    {
      "human_part": "left foot",
      "scene_element": "floor",
      "notes": "the left foot supports the standing pose on the floor"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot supports the standing pose on the floor"
    }
  ],
  "interaction": "A person stands beside the brown small wooden table in a
  ↪  forward-leaning pose. The left hand grips one side of the brown small wooden
  ↪  table, the right hand grips the opposite side, the left foot is planted on the
  ↪  floor, and the right foot is planted on the floor. The person has short black
  ↪  hair, a neutral focused facial expression, a gray shirt, blue jeans, and white
  ↪  sneakers."
}
```

Example 2 input JSON:

```
{
  "interaction": "a person lying on bed with his head on the pillow"
}
```

Example 2 output:

```
{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bed with pillows"
    }
  ],
  "human_part_nodes": [
    "person 1, left leg",
    "person 1, right leg",
    "person 1, head",
    "person 1, hips",
    "person 1, back"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "head",
      "scene_element": "target_object_1",
      "notes": "the head rests on the pillow as part of the bed setup"
    },
    {
      "human_part": "hips",
      "scene_element": "target_object_1",
      "notes": "the hips are supported by the mattress"
    },
    {
      "human_part": "back",
      "scene_element": "target_object_1",
      "notes": "the back rests against the mattress"
    },
    {
```

```
      "human_part": "left leg",
      "scene_element": "target_object_1",
      "notes": "the left leg rests on the bed"
    },
    {
      "human_part": "right leg",
      "scene_element": "target_object_1",
      "notes": "the right leg rests on the bed"
    }
  ],
  "interaction": "A person lies on the bed with pillows in a relaxed static pose. The
  ↪  head rests on a pillow, the back rests on the mattress, the hips are supported
  ↪  by the mattress, the left leg rests on the bed, and the right leg rests on the
  ↪  bed. The person has short black hair, a calm sleepy facial expression, a gray
  ↪  shirt, soft pants, and socks."
}

Example 3 input JSON:

{
  "interaction": "a person stepping over the bathtub wall, one foot inside the tub,
  ↪  the other foot still on the floor, one hand holding the curtain rail for balance"
}

Example 3 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "bathtub"
    },
    {
      "id": "target_object_2",
      "label": "curtain rail"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, left foot",
    "person 1, right foot"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_2",
      "notes": "the left hand grips the curtain rail for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "target_object_1",
      "notes": "the left foot presses inside the bathtub while stepping over the wall"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot stays planted on the floor outside the bathtub"
    }
```

```
  ],
  "interaction": "A person steps over the bathtub wall in a careful balancing pose.
  ↪  The left hand grips the curtain rail, the left foot presses inside the bathtub,
  ↪  and the right foot is planted on the floor outside the tub. The person has short
  ↪  black hair, a focused facial expression, a gray shirt, rolled-up pants, and bare
  ↪  feet."
}

Example 4 input JSON:

{
  "interaction": "a person bending down in front of the lower drawer, one hand pulling
  ↪  the cabinet door open and the other hand holding the counter edge for balance"
}

Example 4 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "lower cabinet door"
    },
    {
      "id": "target_object_2",
      "label": "counter edge"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand",
    "person 1, left foot",
    "person 1, right foot",
    "person 1, hips"
  ],
  "scene_nodes": [
    "target_object_1",
    "target_object_2",
    "floor"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand pulls the lower cabinet door"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_2",
      "notes": "the right hand holds the counter edge for balance"
    },
    {
      "human_part": "left foot",
      "scene_element": "floor",
      "notes": "the left foot supports the squat on the floor"
    },
    {
      "human_part": "right foot",
      "scene_element": "floor",
      "notes": "the right foot supports the squat on the floor"
    }
  ],
```

```
  "interaction": "A person bends down in front of the lower cabinet door in a balanced
  ↪  pose. The left hand pulls the lower cabinet door, the right hand holds the
  ↪  counter edge, the left foot is planted on the floor, and the right foot is
  ↪  planted on the floor. The person has short black hair, a neutral concentrated
  ↪  facial expression, a gray shirt, blue jeans, and white sneakers."
}

Example 5 input JSON:

{
  "interaction": "a person hanging from the pull-up bar with both hands, elbows bent,
  ↪  knees tucked upward, and both feet off the floor"
}

Example 5 output:

{
  "target_objects": [
    {
      "id": "target_object_1",
      "label": "pull-up bar"
    }
  ],
  "human_part_nodes": [
    "person 1, left hand",
    "person 1, right hand"
  ],
  "scene_nodes": [
    "target_object_1"
  ],
  "interaction_edges": [
    {
      "human_part": "left hand",
      "scene_element": "target_object_1",
      "notes": "the left hand grips the pull-up bar"
    },
    {
      "human_part": "right hand",
      "scene_element": "target_object_1",
      "notes": "the right hand grips the pull-up bar"
    }
  ],
  "interaction": "A person hangs from the pull-up bar with elbows bent and knees
  ↪  tucked upward. The left hand grips the pull-up bar, the right hand grips the
  ↪  pull-up bar, the left foot stays off the floor, and the right foot stays off the
  ↪  floor. The person has short black hair, a strained focused facial expression, a
  ↪  gray athletic shirt, dark shorts, and sneakers."
}
```

A.1.2 Human-Frame Inpainting System Prompt Π_{H}

```
Input Image:

- The image is the original scene. Use the interaction description to identify the
↪  existing
target object that must remain fixed and contacted.

Scene Preservation Instructions:

- Use the provided scene image as a fixed camera reference.
```

```
- Insert exactly one person performing the described interaction with the target
↪  object.
- This frame must represent the pre-motion start state: the person is already in
↪  natural contact
with the target object of the scene, but no visible object motion has begun yet.
- Target object is already present in the scene. Do not add it.
- Preserve the camera pose, field of view, perspective, room geometry, lighting,
↪  shadows, textures,
and all existing object identities.
- Do not add new furniture or remove existing items.
- Keep the target object in exactly the same pose, position, orientation, scale,
↪  material,
and appearance as in the original scene.
- Do not move, rotate, deform, lift, translate, or reposition the target object in any
↪  way.
- Only add the human in plausible contact, as if about to start the action in the next
↪  frame.

Output:

- Return only the edited scene image.
```

A.1.3 Best Human-Frame Selection System Prompt Π_{sel}

```
PLACEHOLDER: Final best-frame selection system prompt not fixed yet.

Intended role:
- Given the original scene image, the interaction description or SIG description, and
↪  multiple human-inpainted candidate frames, select the single candidate used as the
↪  visual prior for GVHMR initialization and contact reasoning.

Intended ranking criteria:
1. Action expression: the human should clearly express the requested interaction or
↪  contact-ready pre-action state.
2. Target-object preservation: the target object should remain fixed in pose,
↪  location, scale, material, and appearance relative to the original scene.
3. Plausible body pose: the inserted human should have a physically plausible
↪  full-body pose, orientation, scale, and support.
4. Contact readiness: the relevant body parts should appear close to or in natural
↪  contact with the intended target object surfaces or floor support regions.

Intended output:
- Return the index or identifier of the best candidate and a concise reason for the
↪  selection.
```

A.1.4 Agentic Contact-Overlay System Prompt Π_{con}

```
Provided Images:

Reference Image: An image showing a human interacting with the target object. Use this
↪  image only to infer which human body parts are physically contacting the target
↪  object and which object-surface regions are being touched.

Canvas Image: An image of the same scene without the human. Use this image as the base
↪  image and as the authoritative reference for the target object's final position,
↪  pose, geometry, and boundaries.
```

```
Task Description:

Generate an edited copy of the Canvas Image with solid-color segmentation masks
↪  overlaid only on the target object surface regions that are contacted by the human
↪  in the Reference Image.

Infer the contacted region relative to the target object in the Reference Image, then
↪  place the corresponding contact mask on the matching object region in the Canvas
↪  Image.

Critical Base-Image Rule:
Use the Canvas Image as the only valid base image. Preserve the Canvas Image exactly
↪  and only add the specified solid-color contact masks. Do not use the Reference
↪  Image as the base. Do not redraw, regenerate, stylize, relight, crop, pad, resize,
↪  move, or otherwise alter the Canvas Image.

Analysis Requirements:

1. Contact Inference:
   Analyze the Reference Image to determine which specified human body parts are
   ↪  physically touching the target object.

2. Object-Region Identification:
   For each contacting body part, identify the specific region of the target object
   ↪  being touched, such as the tabletop front edge, tabletop side edge, tabletop
   ↪  upper surface, tabletop underside, table leg, chair seat, chair arm, cabinet
   ↪  handle, ladder rung, rail, shelf, handle, edge, or other relevant object part.

3. Object-Relative Localization:
   Map each inferred contact region onto the corresponding part of the target object
   ↪  in the Canvas Image. Use the Canvas Image object pose and boundaries as ground
   ↪  truth. Prioritize correct placement on the object in the Canvas Image over
   ↪  exact pixel alignment with the Reference Image.

4. Contact Footprint Generation:
   Create a localized contact mask only on the object surface where contact occurs.
   ↪  The mask should represent the apparent contact footprint on the object, not the
   ↪  full visible human body part.

Execution Requirements:

Base Image:
Return the Canvas Image with the contact masks overlaid on top of it.

Canvas Size:
Match the Canvas Image dimensions and coordinate system.

Mask Placement:
Place each mask on the corresponding target-object region in the Canvas Image, using
↪  object-relative correspondence rather than direct coordinate transfer.

Mask Shape:
Use precise, dense, solid-color segmentation masks shaped like the localized contact
↪  footprint on the object surface. The mask should reflect the approximate size,
↪  orientation, and extent of the contact implied by the touching body part.

Object-Surface Clipping:
All mask pixels must be clipped strictly to the target object boundary in the Canvas
↪  Image. No mask may extend onto the human body, floor, wall, background, or any
↪  non-target-object region.

Scene Preservation:
```

```
Use the Canvas Image as a fixed camera reference. Preserve the camera pose, field of
↪  view, perspective, resolution, aspect ratio, room geometry, lighting, shadows,
↪  textures, and all existing object identities.
Keep the target object in exactly the same pose, position, orientation, scale,
↪  material, and appearance as in the Canvas Image. Do not move, rotate, deform,
↪  lift, translate, resize, crop, pad, or reposition the target object or any other
↪  scene element in any way.
Only add the specified solid-color contact mask overlays. Leave every non-overlay
↪  pixel visually unchanged from the Canvas Image.

Colors:
Use only the exact specified solid colors for the contacting body-part masks. Do not
↪  use transparency, antialiasing gradients, shadows, blended colors, or pastel
↪  variants.

Restrictions:
Do not add the generated human body.
Do not copy or paste the full visible hand, foot, hip, arm, leg, or other body-part
↪  silhouette.
Do not transfer masks by raw Reference Image pixel coordinates.
Do not add text labels, arrows, legends, boxes, keypoints, outlines, or any other
↪  annotations.
Show only the original Canvas Image with solid localized contact masks overlaid on the
↪  target object.

Example Guidance:
If the Reference Image shows a person gripping a ladder, infer which rungs or rails
↪  the hands and feet contact. Then place solid masks on those same ladder regions in
↪  the Canvas Image, aligned to the ladder as it appears in the Canvas Image.
```

A.1.5 Contact VLM Evaluation System Prompt Π_{eval}

```
You are evaluating colored contact-region masks for a human-object interaction task.

You are given three images:

Image 1: Reference Image.
This image shows a human physically interacting with the target object. Use it to
↪  infer which body parts are touching which object surfaces.

Image 2: Original Canvas Image.
This image shows the same scene without the human. This is the authoritative geometry
↪  and base image.

Image 3: Current Composite Image.
This is the original Canvas image with colored contact masks pasted onto it. Evaluate
↪  only the colored masks.

Target object:
{target_object}

Color mapping:
{color_mapping}

Task:
Decide whether the colored masks in the Current Composite Image correctly mark the
↪  localized contact regions implied by the Reference Image.

Focus especially on location. For each mask, check whether it is on the correct object
↪  part and correct local surface.
```

````
Evaluate these possible errors:
1. Missing contact: a body part touches the target object in the Reference Image but its mask is absent.
2. Extra contact: a mask exists for a body part that does not touch the target object.
3. Wrong color: a contact uses the wrong color for the body part.
4. Wrong object surface: the mask is on the wrong object part, such as wrong rung, wrong rail, wrong edge, wrong handle, wrong shelf, wrong tabletop region, etc.
5. Shifted mask: the mask is too far left, right, up, or down from the correct contact location.
6. Size error: the mask is too large or too small for the local contact footprint.

For every problem, write a concrete correction instruction for the next ChatGPT image-generation prompt.

Good correction examples:
- "Keep the original canvas unchanged and move only the green right-foot mask upward onto the lower ladder rung."
- "The yellow left-hand mask is missing. Add a small yellow mask on the vertical side rail where the left hand contacts the ladder."
- "The blue foot mask is on the wrong rung. Move it down to the rung directly under the visible foot contact."
- "The red right-hand mask is too large. Keep its center but make it smaller and more focused."
- "This mask is correct. Keep it unchanged."

Bad correction examples:
- "The result is bad."
- "Fix the mask."
- "Try again."

Return strictly valid JSON only:

```json
{
  "done": false,
  "confidence": 0.0,
  "correction_instruction": "text"
}
```

Set "done": true only if all required contacts are present, there are no extra masks, each mask uses the correct color, each mask is on the correct target-object surface, and each mask has acceptable location and size.
````